

ENTERPRISE NETWORK INFRASTRUCTURE BLUEPRINT

FINAL ENGINEERING REPORT

Version 1.0

SYSTEMS ENGINEERING

NETWORK ARCHITECTURE

SECURITY & VALIDATION

PORTFOLIO ENGINEERING BASELINE

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Chapter 1 Executive Summary

1.1 Project Overview

The Enterprise Network Infrastructure Blueprint is a Systems Engineering project that defines, implements, and validates an enterprise campus network architecture for a 150-user, three-floor SME environment.

The project is designed around the operational requirements of ABC Digital Solutions and focuses on four primary engineering concerns: multi-zone enterprise connectivity, Layer 2 resiliency, granular access control, and configuration assurance.

Rather than treating the network as a standalone Packet Tracer topology, the project applies a traceable engineering lifecycle that connects business requirements to formal system requirements, architecture decisions, network design, implementation evidence, validation results, and change-management records.

1.2 Engineering Approach

The project follows a requirements-driven Systems Engineering workflow.

Business and market evidence is decomposed into formal Functional (FR), Security (SR), Availability (AR), Scalability (SC), and Manageability (MG) requirements. These requirements are subsequently traced to architecture decisions, network components, configuration baselines, verification evidence, and acceptance tests.

The resulting engineering chain is maintained as a bidirectional traceability path:

**Business Requirements → System Requirements → Architecture →
Implementation → Validation → Governance**

This approach is intended to demonstrate not only the ability to configure network infrastructure, but also the ability to justify, document, verify, and govern engineering decisions.

The engineering evidence presented throughout this report combines architecture diagrams, implementation records, configuration evidence, validation matrices, and controlled failure-testing results. These artifacts are used to support the traceability and verification claims made in each engineering stage.

1.3 Architecture Baseline

The Version 1 architecture implements a multi-tier Cisco-based campus network with nine production VLANs and a dedicated Native/Dead VLAN 999. Layer 3 inter-VLAN forwarding is

centralized on the Catalyst 3650 CORE-L3-01 using Switched Virtual Interfaces (SVIs).

The access layer uses LACP EtherChannel bundles to provide physical link resiliency, while Rapid-PVST+ provides deterministic Layer 2 convergence. Edge security controls include Port-Fast, BPDU Guard, DHCP Snooping, Sticky Port Security, and Extended ACL policies for inter-zone traffic control.

The architecture also incorporates an edge firewall for perimeter security, NAT-based Internet egress, and controlled access between enterprise security zones.

1.4 Validation Outcome

The implemented infrastructure underwent a twelve-phase acceptance and failure-validation cycle in Cisco Packet Tracer.

The overall validation result was a **94.4% pass rate**. The validation program covered topology and IP-plan verification, trunking, EtherChannel operation, Rapid-PVST+ convergence, Layer 3 routing, security policy behavior, and failure-injection scenarios.

The resilience tests demonstrated sub-two-second Spanning Tree re-convergence following a core-link disruption and maintained EtherChannel operation when an individual member link was removed.

The validation process also identified simulator-specific limitations. In particular, the Packet Tracer Catalyst 3650 parser exhibited limitations when applying Extended ACL bindings to SVI ingress. The policy logic itself was validated, while the simulator constraint was explicitly recorded rather than represented as a successful production-equivalent result.

1.5 Engineering Boundary

Version 1 intentionally does not implement active-active Layer 3 gateway redundancy.

CORE-L3-01 operates as the active Layer 3 gateway, while CORE-L3-02 provides Layer 2 secondary bridging and path redundancy. Consequently, failure of CORE-L3-01 temporarily interrupts inter-VLAN routing.

This behavior is treated as an accepted Version 1 architectural limitation rather than a failed implementation. First-Hop Redundancy Protocol deployment using HSRP or VRRP is explicitly retained as a Version 2 evolution item.

The project therefore distinguishes between validated capabilities, accepted architectural limitations, and simulator constraints.

1.6 Governance and Change Management

The project extends beyond technical configuration by incorporating formal change-management evidence.

During post-implementation verification, a physical endpoint allocation deviation involving

DEV-PC-01 was identified. The endpoint was associated with the wrong physical access-layer location, creating a configuration and port-security consistency issue.

The deviation was processed through **CHANGE-001**, including scope control, stale MAC-address reconciliation, controlled physical reallocation, connectivity verification, and documentation updates.

This provides an auditable example of how an infrastructure deviation can be handled as a controlled engineering change rather than simply being corrected informally.

1.7 Current Baseline Status

At the time of this report, the Version 1 engineering baseline has completed its major architecture, implementation, validation, and documentation activities.

The repository is undergoing a final cross-document data-consistency review before the baseline is formally frozen. This review focuses on maintaining consistency between requirement identifiers, architecture documentation, validation results, and implementation evidence.

The final objective is to ensure that the repository and this report present a single, internally consistent engineering baseline.

Chapter 2 Business Context & Problem Definition

2.1 Organization and Operating Context

The Enterprise Network Infrastructure Blueprint is developed for a representative small-to-medium enterprise environment, referred to as ABC Digital Solutions.

The target organization operates from a three-floor office environment with approximately 150 users. Its workforce includes multiple functional departments with different levels of access sensitivity and different connectivity requirements.

The network therefore needs to support normal business communication while maintaining logical separation between departments, guest users, servers, management traffic, and other infrastructure services.

2.2 Business Environment

The target environment represents a typical enterprise campus network rather than a single flat office LAN.

Users require reliable access to shared infrastructure and Internet services, while security-sensitive departments require controlled communication boundaries. The environment must also support centrally managed network infrastructure and provide sufficient capacity for future organizational growth.

From an engineering perspective, these requirements create several interdependent concerns:

- reliable connectivity between business functions;
- logical segmentation of users and infrastructure;
- controlled communication between security zones;
- protection against common Layer 2 network attacks and configuration errors;
- resilient physical connectivity;
- centralized management and operational visibility; and
- sufficient scalability for future expansion.

2.3 Workforce and Network Demand

The target population of approximately 150 users is distributed across the three-floor office environment.

Different user groups do not require identical network privileges. Consequently, the design uses separate logical network segments rather than treating all endpoints as members of a single broadcast domain.

The resulting requirements call for logical separation between departmental, infrastructure, wireless, and guest traffic. The specific VLAN, IP addressing, and enforcement design used to satisfy these requirements is defined in the subsequent network-design chapters.

The network must additionally accommodate wireless clients, infrastructure management traffic, server-facing connectivity, and guest access.

2.4 Business Problems and Engineering Mapping

The project addresses a set of common infrastructure risks associated with an inadequately segmented enterprise LAN.

2.4.1 Flat Network Exposure

A flat Layer 2 network would allow a compromised endpoint to communicate with a significantly larger portion of the internal environment. Departmental segmentation and controlled inter-zone routing are therefore required to reduce unnecessary lateral communication.

2.4.2 Layer 2 Failure and Configuration Risk

Physical link failures, switching loops, and incorrect access-port configuration can interrupt business connectivity.

The design therefore requires deterministic Layer 2 behavior, redundant physical paths, and edge protection mechanisms capable of limiting the impact of common configuration and topology errors.

2.4.3 Unauthorized or Uncontrolled Access

Guest and departmental traffic must not automatically inherit the same privileges as internal infrastructure.

The network therefore requires explicit access-control policies and security boundaries that can be validated against defined traffic requirements.

2.4.4 Operational Inconsistency

Network infrastructure becomes difficult to maintain when configuration decisions are not documented or traceable to requirements.

The project addresses this problem through standardized configuration baselines, explicit requirement identifiers, architecture decisions, validation evidence, and change-management records.

Table 2.1: Business problem to engineering concern mapping

Business Problem	Engineering Concern	Design Response
Flat network exposure	Lateral movement risk	VLAN segmentation and inter-VLAN ACLs
Layer 2 failure risk	Network downtime	LACP bundling and Rapid-PVST+ root control
Uncontrolled access	Unauthorized resource use	Port Security, DHCP Snooping, and boundary ACLs
Operational inconsistency	Uncontrolled drift	Traceability matrix, configuration baseline, and Change Management

2.5 Engineering Objectives

Based on the identified business context and operational problems, the project establishes the following high-level engineering objectives:

1. Provide segmented and controlled connectivity for the enterprise user population.
2. Establish a structured Layer 2 and Layer 3 campus architecture.
3. Apply security controls at appropriate network enforcement points.
4. Provide physical and Layer 2 resiliency for critical paths.
5. Establish repeatable configuration and validation practices.
6. Maintain traceability from business requirements through implementation and verification.
7. Document known limitations and future architectural evolution explicitly.

These objectives are subsequently decomposed into formal Functional, Security, Availability, Scalability, and Manageability Requirements in Chapter 3.

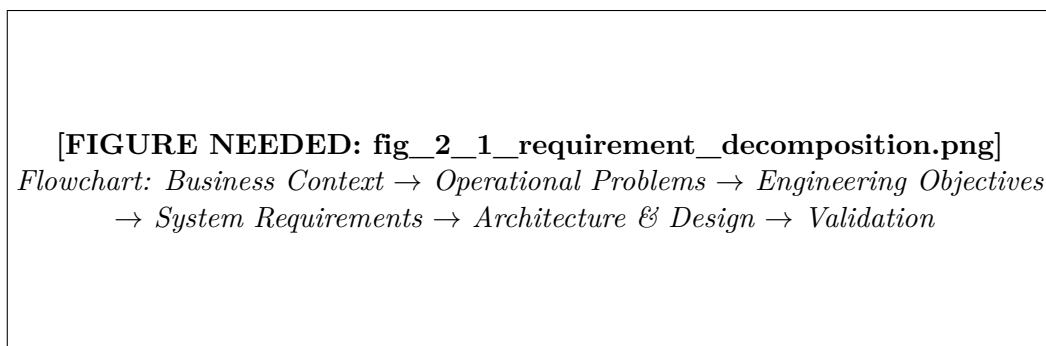


Figure 2.1: Business-to-engineering requirement decomposition

2.6 Project Scope

Version 1 covers the design and validation of the enterprise campus network within a Cisco Packet Tracer environment.

The scope includes:

- enterprise VLAN and IP addressing design;
- Layer 2 switching and Layer 3 SVI routing;
- LACP EtherChannel;
- Rapid-PVST+ spanning-tree operation;
- Layer 2 access security controls;
- Extended ACL-based traffic control;
- perimeter firewall and NAT functions;
- infrastructure management considerations;
- validation and failure-injection testing; and
- controlled change-management documentation.

The project does not claim to reproduce every capability of a production enterprise environment. In particular, Version 1 does not implement active-active Layer 3 gateway redundancy. First-Hop Redundancy Protocol deployment is retained as a future evolution item.

2.7 Success Criteria

The project is considered successful when the defined requirements can be traced to concrete architecture or configuration decisions and the resulting implementation can be verified through documented test evidence.

Success is therefore evaluated through more than simple network reachability. The baseline must demonstrate:

- requirement-to-design traceability;
- correct implementation of the defined network architecture;
- enforcement of intended security boundaries;
- successful operation under defined failure scenarios;
- documented handling of implementation deviations; and
- explicit identification of simulator and architectural limitations.

These criteria establish the basis for the formal requirements and validation framework presented in the following chapters.

Chapter 3 Requirements

This chapter defines the engineering requirements for the Enterprise Network Infrastructure Blueprint.

The requirements establish the measurable baseline against which the architecture, network design, implementation, security controls, and validation activities are evaluated.

The Version 1 requirement model is divided into five requirement categories:

- Functional Requirements (FR)
- Security Requirements (SR)
- Availability Requirements (AR)
- Scalability Requirements (SC)
- Manageability Requirements (MG)

This classification provides the traceability foundation used throughout the remainder of the engineering report.

3.1 Functional Requirements

Functional requirements define the services and network behaviors that the enterprise infrastructure must provide.

Table 3.1: Functional requirements

ID	Requirement	Engineering Definition
FR-01	Inter-Departmental Communication	The network shall provide controlled Layer 3 communication between authorized departmental VLANs.
FR-02	Guest Internet Egress	The guest network shall provide Internet-oriented connectivity while remaining logically isolated from protected internal segments.
FR-03	Automated IP Allocation	Client endpoints shall receive IP configuration through the defined DHCP architecture.

ID	Requirement	Engineering Definition
FR-04	Wireless Mobility Support	The network shall support wireless endpoint connectivity through the defined access-layer architecture.
FR-05	Secure Remote Administration	Network infrastructure shall support controlled administrative access using the defined secure management approach.

3.2 Security Requirements

Security requirements define the controls used to reduce unauthorized access, Layer 2 attacks, network misconfiguration, and unintended traffic exposure.

Table 3.2: Security requirements

ID	Requirement	Engineering Definition
SR-01	Subnet Logical Isolation	Departmental and functional network segments shall be separated through VLAN and Layer 3 boundaries.
SR-02	Confidential Asset Protection	Sensitive infrastructure and management resources shall be protected from unauthorized endpoint access.
SR-03	Guest Untrusted Isolation	Guest endpoints shall remain isolated from protected internal resources.
SR-04	Rogue DHCP Mitigation	The access layer shall include controls to prevent unauthorized DHCP services from influencing client configuration.
SR-05	Layer 2 Loop Suppression	Spanning-tree controls shall prevent persistent Layer 2 forwarding loops.
SR-06	Control Plane Hardening	Infrastructure devices shall apply appropriate control-plane and management-plane hardening measures.

3.3 Availability Requirements

Availability requirements define the mechanisms used to maintain network connectivity when individual links or forwarding paths experience failure.

Table 3.3: Availability requirements

ID	Requirement	Engineering Definition
AR-01	Uplink Redundancy and Load Sharing	Critical switch uplinks shall use link aggregation where supported to provide additional capacity and resilience.
AR-02	Spanning-Tree and Path Redundancy	The switching topology shall maintain redundant physical paths while using spanning-tree controls to prevent loops.
AR-03	Inline Power Delivery	The access-layer architecture shall support inline power delivery for appropriate powered endpoints.

3.4 Scalability Requirements

Scalability requirements define how the Version 1 architecture can accommodate additional users, endpoints, and network segments without requiring a fundamental redesign.

Table 3.4: Scalability requirements

ID	Requirement	Engineering Definition
SC-01	Modular Subnet Expansion	The addressing architecture shall allow additional network segments to be introduced without redesigning the complete addressing model.
SC-02	Floor Density Replication	The access-layer design shall support replication across additional physical areas or floors using the established design pattern.
SC-03	IP Capacity Reservation	Addressing shall provide sufficient reserved capacity for expected endpoint growth within the Version 1 engineering scope.

3.5 Manageability Requirements

Manageability requirements establish the engineering practices needed to maintain consistent configuration, traceability, and recoverability.

Table 3.5: Manageability requirements

ID	Requirement	Engineering Definition
MG-01	Standardized Configuration Codebases	Device configurations shall follow a consistent and reviewable configuration structure.
MG-02	Deterministic Addressing Scheme	Network addressing shall follow a predictable and documented allocation model.
MG-03	Logical-to-Physical Traceability	Logical network segments shall be traceable to their physical implementation and associated infrastructure.
MG-04	Disaster Recovery and Configuration Backups	Critical device configurations shall be preserved to support recovery and controlled restoration.

3.6 Requirement Prioritization

The requirements form a single engineering baseline rather than a collection of independent feature requests.

Functional requirements establish the required network services. Security requirements define the protection boundaries around those services. Availability requirements address resilience of the forwarding infrastructure, while scalability and manageability requirements establish the conditions necessary for sustainable operation.

The resulting dependency can be summarized as:

**Functional Services → Security Controls → Availability → Scalability →
Manageability**

This structure is carried forward into the architecture and validation phases.

3.7 Requirement Traceability

Each requirement is intended to remain traceable through the engineering lifecycle.

The traceability chain is:

**Requirement → Architecture → Network Design → Implementation → Validation
→ Acceptance**

The detailed requirement-to-engineering mapping is provided in **Appendix A – Requirement**

Traceability.

This separation keeps Chapter 3 focused on the engineering intent while Appendix A provides the detailed evidence mapping.

3.8 Version 1 Requirement Boundary

The Version 1 baseline deliberately distinguishes implemented requirements from future evolution.

In particular, Layer 2 and link-level redundancy are implemented through the switching architecture, LACP, and spanning-tree design. Active Layer 3 gateway redundancy is not part of the Version 1 implementation and is reserved for future evolution.

This distinction is important for acceptance: a deferred capability must not be represented as an implemented Version 1 requirement.

The final requirement status is therefore determined through the validation and acceptance process documented in Chapters 8 and 13.

Chapter 4 Architecture

This chapter defines the Version 1 enterprise network architecture and establishes the structural relationship between the perimeter, core, access, and endpoint layers.

The architecture is represented through five complementary views:

- logical topology;
- physical infrastructure layout;
- Layer 3 routing flow;
- security boundary model; and
- LACP EtherChannel resilience architecture.

These views describe the architectural structure of the system. Detailed addressing, VLAN allocation, interface assignments, and device-level implementation are specified in Chapter 5.

4.1 Architectural Principles

The Version 1 architecture follows a layered enterprise campus model.

The design separates the infrastructure into four principal tiers:

1. Security / WAN Edge;
2. Core / Layer 3 Distribution;
3. Access Layer; and
4. Representative Endpoints.

This separation provides clear functional boundaries between perimeter connectivity, centralized Layer 3 forwarding, endpoint access, and user or infrastructure devices.

The architecture also separates logical security boundaries from physical connectivity. VLAN segmentation and Layer 3 interfaces define logical boundaries, while LACP and spanning-tree mechanisms provide resilience at the switching layer.

4.2 Logical Topology

The logical topology represents the primary traffic and control relationships between the major infrastructure tiers.

The architecture is organized into four logical tiers.

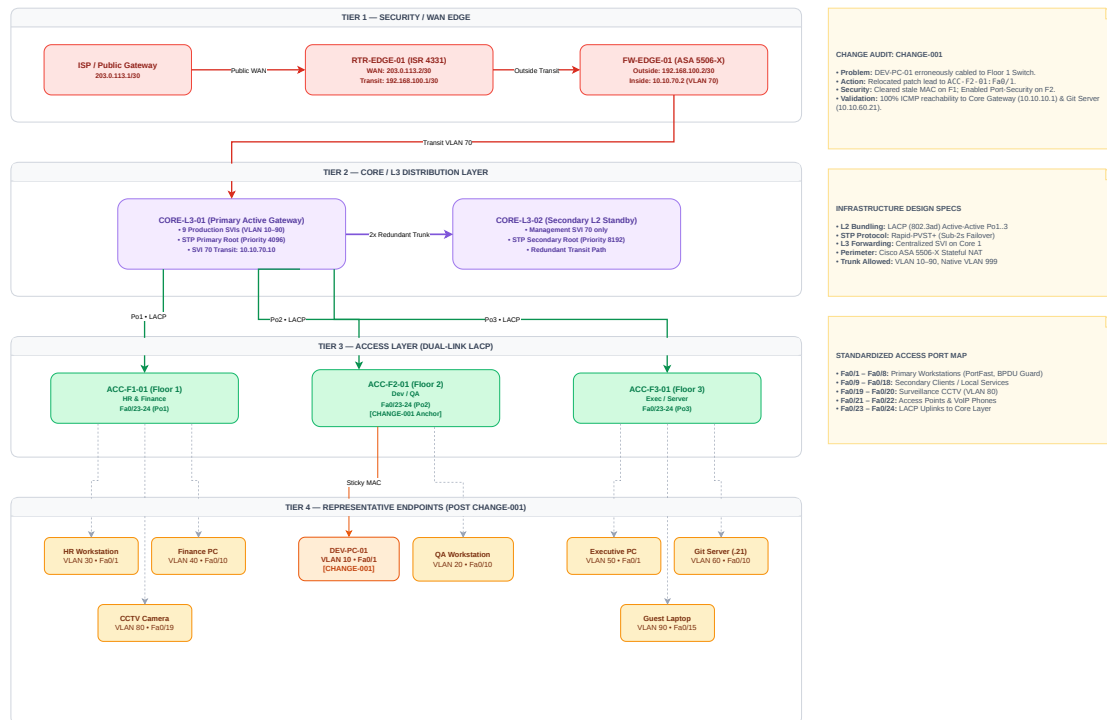


Figure 4.1: Version 1 logical network architecture

4.2.1 Security and WAN Edge

The perimeter consists of the ISP/public gateway, the edge router RTR-EDGE-01, and the firewall FW-EDGE-01.

The edge router provides the WAN-facing and internal transit connectivity, while the Cisco ASA 5506-X provides the defined stateful perimeter security and NAT boundary.

The logical topology identifies the following principal transit addresses:

- WAN connection: 203.0.113.0/30;
- router-to-firewall transit: 192.168.100.0/30;
- firewall inside interface: VLAN 70;
- core management/transit segment: VLAN 70.

The perimeter therefore provides a distinct boundary between the external network and the internal campus architecture.

4.2.2 Core and Layer 3 Distribution

The core consists of two Catalyst 3650 switches:

- CORE-L3-01;
- CORE-L3-02.

CORE-L3-01 is the primary active Layer 3 gateway. It hosts the production SVIs and provides centralized inter-VLAN forwarding.

The logical architecture identifies CORE-L3-01 as the primary active gateway and CORE-L3-02 as the secondary Layer 2 standby device.

This distinction is intentional. Version 1 does not implement an active first-hop redundancy protocol such as HSRP or VRRP. Therefore, CORE-L3-02 must not be interpreted as an active redundant Layer 3 gateway.

Instead, it contributes to the switching topology through its Layer 2 role and redundant transit path.

4.2.3 Access Layer

The access layer contains three floor-specific switches:

- ACC-F1-01;
- ACC-F2-01;
- ACC-F3-01.

Each access switch represents a repeatable floor-level deployment pattern.

The access layer connects upward to the core using dual-link LACP bundles. This provides link aggregation and resilience while maintaining a controlled Layer 2 topology.

4.2.4 Representative Endpoints

Representative endpoint classes are included to demonstrate how logical VLAN segmentation maps to endpoint connectivity.

The architecture includes examples such as:

- HR workstation;
- Finance workstation;
- development workstation;
- QA workstation;
- executive workstation;
- Git server;
- CCTV camera; and

- guest laptop.

These endpoints are representative rather than an exhaustive inventory of every endpoint in the simulated enterprise.

4.3 Physical Architecture

The physical architecture describes the placement of the principal network infrastructure within the server-room environment.

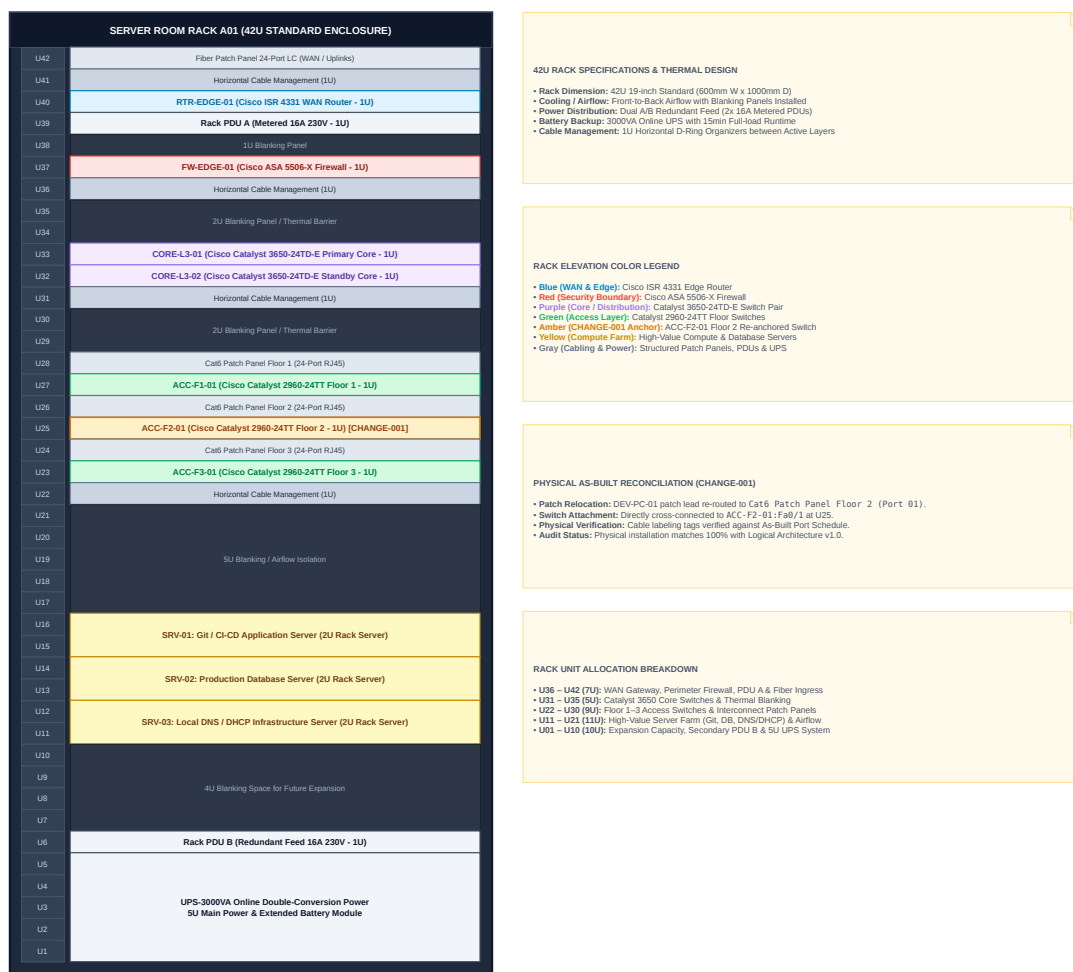


Figure 4.2: Version 1 physical infrastructure rack layout

The physical layout uses a standard 42U rack representation and separates the principal infrastructure components according to their operational roles.

The physical design supports the logical architecture by providing a clear mapping between the perimeter, core switching, access infrastructure, and supporting equipment.

The physical representation is an engineering baseline for the simulated environment rather than a claim that physical hardware has been deployed in a production facility.

4.4 Layer 3 Routing Architecture

Layer 3 forwarding is centralized at the core distribution layer.

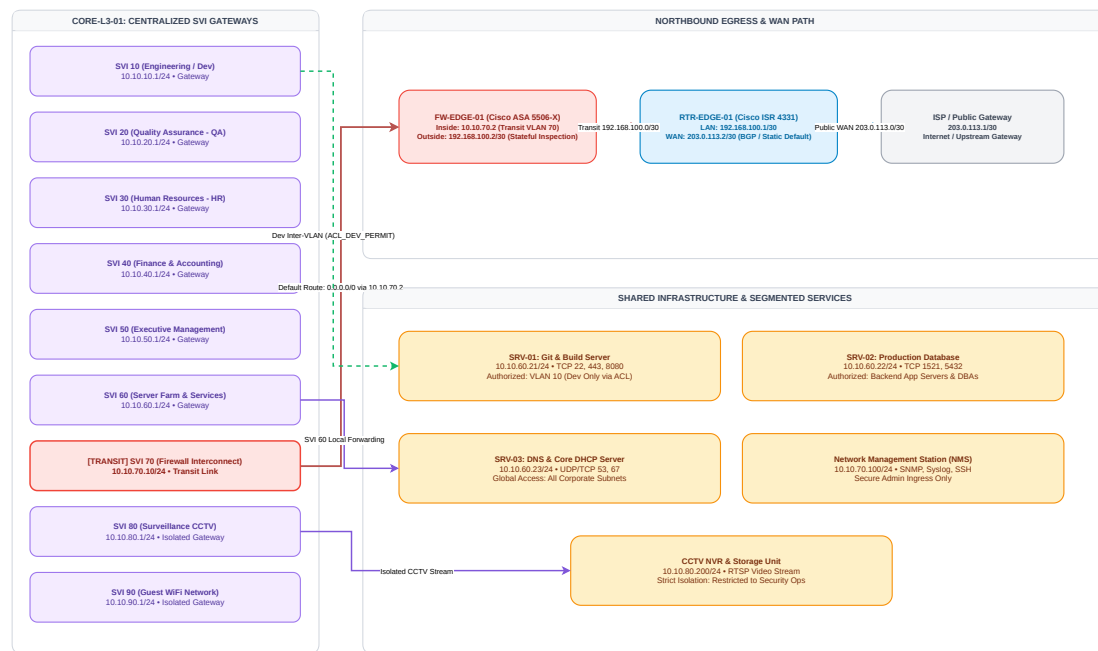


Figure 4.3: Version 1 Layer 3 routing and traffic flow

The routing architecture follows the principle that endpoint traffic is assigned to a VLAN and reaches its corresponding default gateway through the centralized SVI architecture.

CORE-L3-01 provides the active gateway interfaces for the production VLANs and performs inter-VLAN routing.

The access switches therefore primarily provide Layer 2 endpoint connectivity and transport VLAN traffic toward the core.

This design keeps the Layer 3 boundary centralized and makes the routing model deterministic for validation.

4.4.1 Gateway and Transit Model

VLAN 70 serves as the management/transit segment associated with the infrastructure and perimeter connectivity.

The core architecture identifies the principal gateway interface on CORE-L3-01 as:

10.10.70.10

The firewall's internal interface participates in the same architectural boundary, while the router-to-firewall transit uses the dedicated 192.168.100.0/30 network.

This separation avoids conflating the WAN-facing network with the internal enterprise VLAN structure.

4.5 Security Boundary Architecture

Security controls are applied at multiple architectural boundaries.

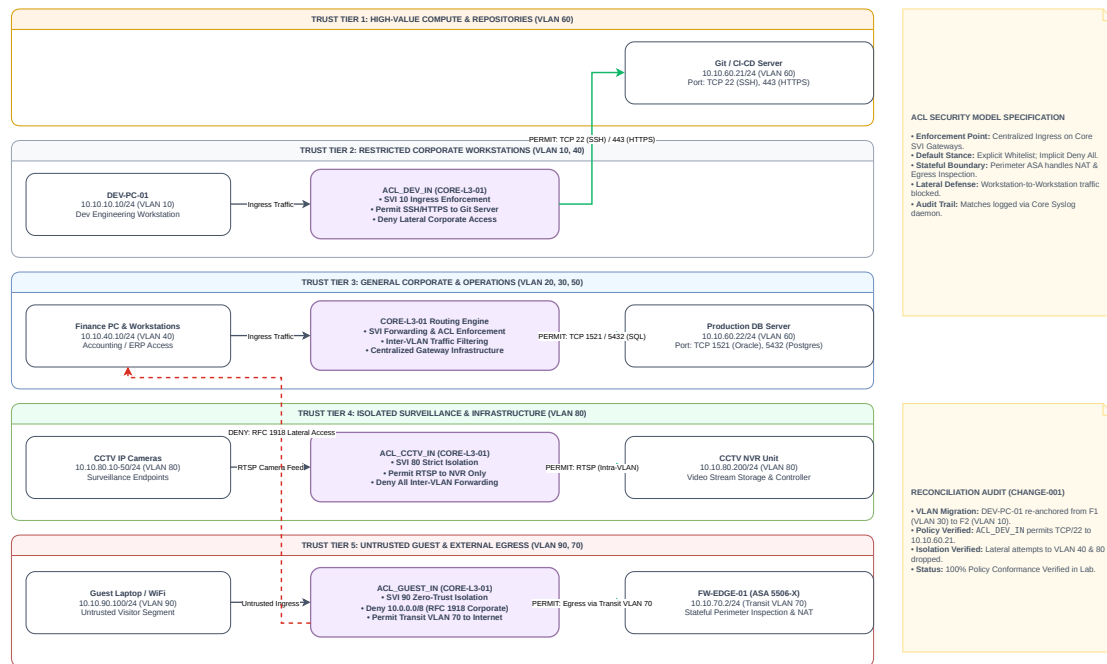


Figure 4.4: Version 1 internal ACL security boundaries

The architecture establishes security boundaries through:

- VLAN segmentation;
- Layer 3 gateway boundaries;
- perimeter firewall controls;
- ACL policy definitions;
- access-layer Layer 2 security controls; and
- isolated guest connectivity.

The security architecture therefore does not depend on a single control.

Instead, protection is distributed across the perimeter, routing boundary, and access layer.

Detailed security implementation and validation are addressed in Chapter 6.

4.6 LACP and Switching Resilience

The access-to-core connectivity uses dual physical links aggregated through LACP.

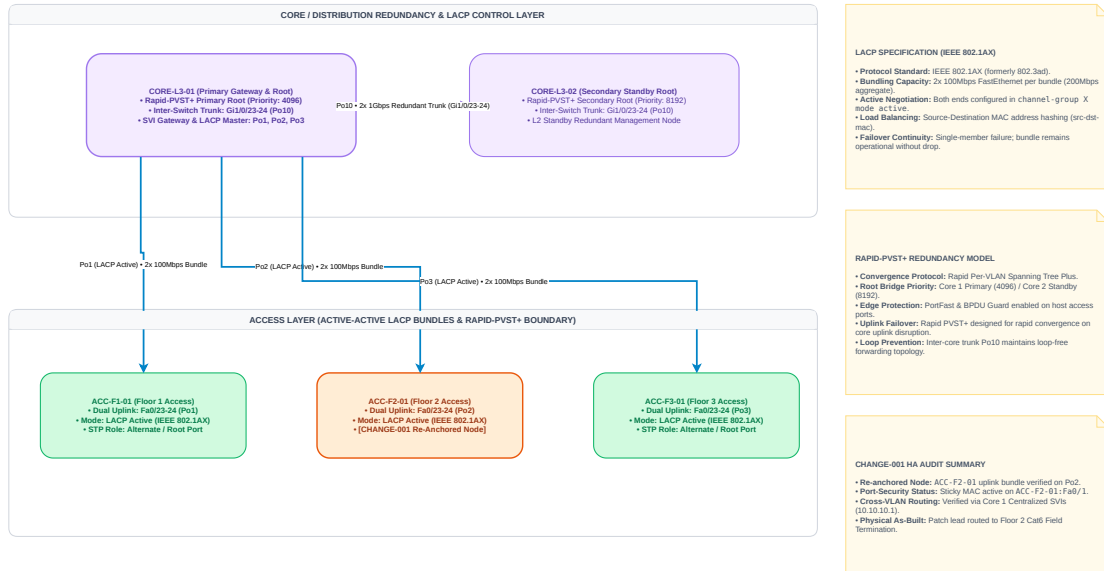


Figure 4.5: Version 1 LACP EtherChannel resilience architecture

The architecture defines three principal access-layer port channels:

- ACC-F1-01: Po1;
- ACC-F2-01: Po2;
- ACC-F3-01: Po3.

Each bundle uses dual physical member links.

LACP provides link-level redundancy and load sharing within the EtherChannel architecture. This mechanism is intentionally distinguished from Layer 3 gateway redundancy.

Spanning-tree controls complement LACP by maintaining a loop-free Layer 2 topology when multiple physical paths exist.

4.7 Spanning-Tree Architecture

The switching architecture uses Rapid-PVST+ to control Layer 2 forwarding paths.

CORE-L3-01 is configured as the primary spanning-tree root, while CORE-L3-02 provides the secondary root position.

This arrangement establishes deterministic Layer 2 path selection while preserving an alternate switching path.

The architecture therefore separates two resilience functions:

- LACP protects and aggregates individual uplink paths;
- Rapid-PVST+ controls redundant Layer 2 forwarding paths.

The resulting design provides Layer 2 resilience without claiming full Layer 3 gateway redundancy.

4.8 Architectural Boundaries

The Version 1 architecture has several deliberate boundaries.

4.8.1 Layer 2 / Layer 3 Boundary

Endpoint VLANs terminate at the centralized SVI architecture on CORE-L3-01.

The access layer does not independently perform the primary inter-VLAN routing function.

4.8.2 Internal / External Boundary

The firewall and edge routing infrastructure establish the boundary between the internal enterprise network and the external WAN.

4.8.3 Trusted / Untrusted Boundary

Guest connectivity is treated as an untrusted logical segment and is separated from protected internal resources through the defined security architecture.

4.8.4 Active / Standby Boundary

The two core switches do not provide active-active Layer 3 gateway redundancy in Version 1.

CORE-L3-01 is the active Layer 3 gateway, while CORE-L3-02 provides Layer 2 standby and redundant switching-path functionality.

This boundary is an explicit Version 1 limitation and future evolution point.

4.9 Architecture Traceability

The architecture translates the requirements established in Chapter 3 into a physical and logical network structure.

The principal relationships are:

Requirements → Architectural Layers → Network Design → Implementation

Functional requirements are primarily represented through the endpoint, access, and Layer 3 routing architecture.

Security requirements are represented through VLAN segmentation, security boundaries, perimeter controls, ACL policy, and access-layer controls.

Availability requirements are represented through LACP and spanning-tree mechanisms.

Scalability requirements are represented through the repeatable floor-level access design and modular VLAN architecture.

Manageability requirements are represented through standardized device roles, deterministic logical-to-physical relationships, and configuration traceability.

Detailed implementation parameters are intentionally deferred to Chapter 5.

4.10 Architectural Baseline

The Version 1 architectural baseline can therefore be summarized as follows:

- a dedicated security and WAN edge;
- centralized Layer 3 routing at CORE-L3-01;
- a secondary core switch providing Layer 2 redundancy;
- three repeatable floor-level access switches;
- dual-link LACP uplinks;
- Rapid-PVST+ for Layer 2 path control;
- VLAN-based logical segmentation;
- a defined perimeter security boundary; and
- explicit separation between implemented V1 capabilities and deferred future capabilities.

This architecture forms the structural baseline for the detailed network design presented in Chapter 5 and the security controls presented in Chapter 6.

Chapter 5 Network Design

This chapter translates the architectural baseline defined in Chapter 4 into a concrete network design.

While Chapter 4 describes the structural organization of the enterprise network, this chapter defines the logical segmentation, Layer 3 gateway model, Layer 2 connectivity, routing relationships, and resilience mechanisms used by the Version 1 implementation.

The design is based on the following principles:

- deterministic network segmentation;
- centralized Layer 3 gateway services;
- controlled Layer 2 transport;
- redundant access uplinks;
- predictable spanning-tree behavior; and
- clear logical-to-physical traceability.

5.1 Network Segmentation

The network uses VLAN-based segmentation to separate departmental, infrastructure, service, and guest traffic.

Each logical network segment is associated with a dedicated Layer 3 gateway interface on the active core switch.

The segmentation model provides a deterministic boundary between different classes of endpoints while allowing authorized inter-VLAN communication through the centralized Layer 3 routing architecture.

The principal segmentation categories include:

- departmental user networks;
- infrastructure and management networks;
- service/server networks;
- security or monitoring-related segments; and
- guest connectivity.

This design supports the security and functional requirements defined in Chapter 3 without requiring each access switch to perform independent Layer 3 routing.

5.2 VLAN Design

VLANs form the primary Layer 2 segmentation mechanism.

The Version 1 architecture includes a dedicated management/transit VLAN 70. The corresponding active gateway on CORE-L3-01 is:

10.10.70.10/24

VLAN 70 provides the logical boundary used for infrastructure management and the associated internal transit relationship.

The remaining production VLANs are terminated through SVIs on CORE-L3-01.

The complete VLAN-to-subnet and interface allocation is treated as the authoritative addressing baseline for implementation and validation. Device-level configuration is documented separately to avoid duplicating configuration evidence throughout this chapter.

5.3 IP Addressing Design

The addressing model follows a deterministic subnet allocation approach.

Each logical network receives its own subnet, with the Layer 3 gateway positioned at the centralized core.

This provides three important properties:

1. predictable default-gateway placement;
2. straightforward troubleshooting; and
3. direct traceability between VLANs, SVIs, and endpoint networks.

The infrastructure also uses dedicated transit networks where required.

The WAN-facing architecture uses:

203.0.113.0/30

The router-to-firewall transit uses:

192.168.100.0/30

These transit networks are kept distinct from the internal production VLAN addressing model.

5.4 Layer 3 Gateway Design

CORE-L3-01 is the active Layer 3 gateway for the Version 1 network.

The switch provides the production SVIs and performs inter-VLAN routing.

The architecture intentionally avoids distributing gateway functions across the access layer.

This creates a centralized Layer 3 boundary:

Endpoint → Access VLAN → CORE-L3-01 SVI → Inter-VLAN Routing

CORE-L3-02 does not provide an active redundant Layer 3 gateway in Version 1.

Its role is limited to Layer 2 switching redundancy and the associated alternate switching path.

Therefore, the design does not claim HSRP or VRRP-based first-hop redundancy.

5.5 Access-Layer Design

The access layer follows a repeatable floor-based design pattern.

The Version 1 deployment contains:

- ACC-F1-01;
- ACC-F2-01; and
- ACC-F3-01.

Each access switch provides endpoint connectivity and transports the required VLANs toward the core.

The access layer is therefore primarily a Layer 2 domain.

This design simplifies endpoint onboarding because the same architectural pattern can be applied to additional floors or network areas without changing the fundamental Layer 3 model.

5.6 Trunk Design

Inter-switch connectivity uses trunk links to transport the required VLANs between the access and core layers.

The trunk design ensures that VLAN membership remains consistent across the Layer 2 path between an access switch and its associated core connectivity.

The logical relationship is:

Endpoint Access Port → Access VLAN → 802.1Q Trunk → Core SVI

Access ports are assigned according to endpoint requirements, while inter-switch links operate as VLAN trunks.

This separation prevents endpoint-facing ports from becoming unintended Layer 2 transit paths.

5.7 LACP EtherChannel Design

The access switches use dual uplinks toward the core.

These physical links are aggregated through LACP to form logical port channels.

The Version 1 architecture identifies the following principal port-channel relationships:

- ACC-F1-01 — Po1;
- ACC-F2-01 — Po2; and
- ACC-F3-01 — Po3.

Each logical bundle is composed of two physical member links.

The resulting design provides:

- link-level redundancy;
- aggregated bandwidth;
- simplified spanning-tree treatment of bundled links; and
- continued connectivity following the loss of an individual member link.

LACP is therefore treated as a link-level availability mechanism, not as Layer 3 gateway redundancy.

5.8 Spanning-Tree Design

Rapid-PVST+ provides Layer 2 loop prevention and deterministic forwarding-path selection.

The Version 1 core hierarchy establishes:

- CORE-L3-01 as the primary spanning-tree root;
- CORE-L3-02 as the secondary spanning-tree root.

This arrangement establishes predictable Layer 2 forwarding behavior while retaining an alternate switching path.

Spanning-tree therefore complements LACP:

LACP → link aggregation and member-link resilience

Rapid-PVST+ → loop prevention and Layer 2 path selection

The two mechanisms address different failure domains and should not be interpreted as interchangeable.

5.9 DHCP Design

Automated IP allocation is part of the functional network design.

Client-facing VLANs use the defined DHCP architecture to provide endpoint addressing without requiring static configuration for ordinary user devices.

The DHCP design is logically separated from the Layer 3 gateway function:

DHCP Service → Client Address Allocation

while:

CORE-L3-01 SVI → Default Gateway

This separation allows endpoint addressing and gateway forwarding to remain independently traceable.

Detailed DHCP configuration and verification are documented in the implementation and validation chapters.

5.10 WAN and Perimeter Connectivity

The external connectivity path follows the architecture defined in Chapter 4.

The principal path is:

WAN → RTR-EDGE-01 → FW-EDGE-01 → Internal Network

The WAN-side addressing baseline uses 203.0.113.0/30.

The router-to-firewall connection uses the dedicated 192.168.100.0/30 transit network.

The firewall then connects toward the internal management/transit architecture through VLAN 70.

This arrangement separates:

- external WAN addressing;
- edge-to-firewall transit addressing; and
- internal enterprise VLAN addressing.

5.11 Logical-to-Physical Mapping

The network design maintains explicit traceability between logical segments and physical infrastructure.

The relationship can be summarized as:

VLAN → Subnet → SVI → Core → Trunk → Access Switch → Endpoint

This mapping is important for both troubleshooting and validation.

An endpoint connectivity problem can therefore be evaluated systematically across the following boundaries:

1. endpoint access port;
2. access VLAN;
3. trunk transport;
4. core SVI;
5. Layer 3 routing; and
6. upstream/perimeter connectivity.

This provides a deterministic troubleshooting model for the enterprise network.

5.12 Design Trade-offs

The Version 1 network design deliberately prioritizes deterministic behavior, validation simplicity, and architectural traceability.

Centralizing Layer 3 gateway services on **CORE-L3-01** simplifies routing and validation but creates a gateway-level availability limitation because **CORE-L3-02** is not an active redundant Layer 3 gateway.

Similarly, LACP provides link-level resilience without providing full device-level gateway redundancy.

The resulting design therefore provides meaningful Layer 2 and link resilience while retaining a clearly documented limitation at the Layer 3 gateway layer.

This trade-off is intentional for Version 1 and is carried forward as a future-evolution item rather than being represented as an implemented capability.

5.13 Network Design Traceability

The network design implements the architectural principles defined in Chapter 4 and provides the technical foundation for the implementation described in Chapter 7.

The relationship is:

Requirements → Architecture → Network Design → Implementation → Validation

Functional requirements are represented through VLAN connectivity, Layer 3 gateway services, DHCP, and endpoint access.

Availability requirements are represented through LACP and Rapid-PVST+.

Scalability requirements are represented through the repeatable floor-based access design and modular network segmentation.

Manageability requirements are represented through deterministic addressing, standardized device roles, and logical-to-physical traceability.

Security requirements are implemented through the network segmentation and boundary model, with detailed controls addressed in Chapter 6.

5.14 Version 1 Network Design Baseline

The Version 1 network design is therefore characterized by:

- VLAN-based network segmentation;
- centralized Layer 3 gateway services on CORE-L3-01;
- Layer 2 redundancy through CORE-L3-02;
- deterministic SVI-based inter-VLAN routing;
- trunk-based VLAN transport;
- LACP EtherChannel uplinks;
- Rapid-PVST+ Layer 2 path control;
- DHCP-based endpoint addressing;
- dedicated WAN and perimeter transit networks; and
- explicit separation between implemented V1 resilience and deferred Layer 3 gateway redundancy.

The resulting network design provides the detailed technical baseline required for implementation and subsequent validation.

Chapter 6 Security

This chapter defines the security architecture and control framework for the Version 1 enterprise network.

The security design follows a defense-in-depth model in which security controls are distributed across the Layer 2 access layer, the centralized Layer 3 boundary, the management plane, and the perimeter.

The security architecture is derived from six System Security Requirements:

- SR-01 — Subnet Logical Isolation;
- SR-02 — Confidential Asset Protection;
- SR-03 — Guest Untrusted Isolation;
- SR-04 — Rogue DHCP Mitigation;
- SR-05 — Layer 2 Loop Suppression; and
- SR-06 — Control Plane Hardening.

The design also follows three principal security principles:

1. least-privilege access;
2. role-based network segmentation; and
3. defense in depth.

6.1 Security Design Principles

The Version 1 security architecture does not depend on a single security mechanism.

Instead, controls are applied at multiple points in the network path.

6.1.1 Least Privilege

Inter-VLAN communication follows a default-deny security model.

Traffic between different security zones is permitted only when required by an explicitly defined business or infrastructure function.

This prevents unrestricted lateral movement between departmental networks.

6.1.2 Role-Based Segmentation

Each major organizational or operational role is assigned to a dedicated logical network.

The resulting segmentation separates user groups, infrastructure, servers, and guests into distinct Layer 2 and Layer 3 boundaries.

6.1.3 Defense in Depth

The design combines:

- Layer 2 edge protection;
- Layer 3 ACL policy;
- management-plane restriction;
- perimeter security; and
- spanning-tree protection.

Failure or bypass of one individual control therefore does not automatically remove every security boundary.

6.2 Threat Model

The security design addresses six primary operational threats.

Table 6.1: Primary security threats and corresponding controls

ID	Threat	Primary Control
THR-01	Unauthorized inter-VLAN lateral access	Inter-VLAN extended ACLs
THR-02	Rogue DHCP server insertion	DHCP Snooping and untrusted access ports
THR-03	MAC spoofing and unauthorized devices	Port Security with sticky MAC learning
THR-04	Unauthorized STP/BPDU injection	BPDU Guard and PortFast
THR-05	Guest lateral intrusion	Guest isolation ACL
THR-06	Unauthorized control-plane probing	Management VLAN 70 restriction

The threat model establishes the relationship between an operational risk and the corresponding enforcement mechanism.

This provides traceability from the security requirements to specific technical controls.

6.3 Security Zones

The network is divided into logical security zones through VLAN and subnet segmentation.

The principal security zones include:

- Development;
- QA;
- HR;
- Finance;
- Management;
- Server infrastructure; and
- Guest.

Each zone represents a distinct trust boundary.

Traffic crossing between zones is routed through the centralized Layer 3 gateway architecture and is therefore subject to the defined security policy.

The design intentionally avoids treating the internal network as a single trusted broadcast domain.

6.4 Inter-VLAN Access Control

The primary Layer 3 security mechanism is the use of extended Access Control Lists on the core SVI interfaces.

The intended policy model is:

Source VLAN → Core SVI → Security Policy → Destination

The ACL design follows a least-privilege model in which explicitly required services are permitted while unnecessary lateral access is denied.

6.4.1 Development Policy

The Development security policy is represented by `ACL_DEV_IN` on `Vlan10`.

The intended policy permits required development services, including:

- SSH/Git traffic using TCP port 22;
- HTTPS using TCP port 443;
- DNS using UDP port 53; and

- required Internet egress.

The policy denies unnecessary access to protected departmental and management networks.

6.4.2 Finance Policy

The Finance security policy is represented by `ACL_FIN_IN` on `Vlan40`.

The intended policy permits required enterprise application services, including:

- Microsoft SQL Server traffic using TCP port 1433;
- HTTPS using TCP port 443;
- DNS using UDP port 53; and
- required Internet egress.

Unnecessary access to Development, QA, and Guest networks is denied.

6.4.3 Guest Isolation Policy

Guest traffic is treated as untrusted.

The intended policy is represented by `ACL_GUEST_IN` on `Vlan90`.

Private RFC 1918 destinations are denied:

- `10.0.0.0/8`;
- `172.16.0.0/12`; and
- `192.168.0.0/16`.

Remaining permitted traffic is intended for external Internet access.

This creates an explicit guest-to-internal security boundary.

6.5 Layer 2 Edge Security

Layer 2 controls are deployed at the access layer to reduce the risk of unauthorized devices and malicious Layer 2 behavior.

6.5.1 DHCP Snooping

DHCP Snooping protects against unauthorized DHCP servers.

The access switches enable DHCP Snooping for the relevant VLANs.

Core-facing uplinks are designated as trusted interfaces, while ordinary user-facing ports remain untrusted.

The intended traffic model is:

Authorized DHCP Server → Trusted Path → Client

Unauthorized DHCP offers arriving from an untrusted endpoint port are therefore rejected by the access-layer security mechanism.

6.5.2 Port Security

User-facing access ports employ Port Security.

The design uses sticky MAC learning with a maximum of two MAC addresses per user-facing port.

The limit accommodates the intended workstation and VoIP pass-through deployment model.

Security violations are handled using the defined restrictive violation behavior rather than allowing unrestricted MAC learning.

6.5.3 BPDU Guard and PortFast

User-facing edge ports use PortFast and BPDU Guard.

PortFast allows an endpoint-facing interface to transition rapidly into the forwarding state.

BPDU Guard protects the spanning-tree topology by treating an unexpected BPDU received on an edge port as a security violation.

This prevents an unauthorized switch from influencing the spanning-tree topology through a user-facing connection.

6.6 Management Plane Protection

Administrative access is isolated through Management VLAN 70.

The management network uses:

10.10.70.0/24

with the active core gateway:

10.10.70.10

Management-plane services such as SSH and SNMP are intended to be accessible only from the authorized management network.

User and guest networks are not treated as trusted administrative sources.

The design therefore establishes a dedicated control-plane boundary between ordinary endpoint traffic and network administration.

6.7 Security Requirements Traceability

The security controls map directly to the canonical System Security Requirements.

Table 6.2: Security requirements and control mapping

ID	Security Objective	Primary Control
SR-01	Subnet Logical Isolation	Inter-VLAN extended ACLs
SR-02	Confidential Asset Protection	Explicit service and destination filtering
SR-03	Guest Untrusted Isolation	Guest isolation ACL blocking private RFC 1918 destinations
SR-04	Rogue DHCP Mitigation	DHCP Snooping with untrusted user ports
SR-05	Layer 2 Loop Suppression	BPDU Guard and PortFast
SR-06	Control Plane Hardening	Management VLAN 70 restriction

This mapping maintains the security portion of the report's requirements-to-implementation golden thread.

6.8 Security Enforcement Boundary

The intended security enforcement model can be summarized as:

Endpoint → Layer 2 Controls → Core SVI → ACL Policy → Perimeter

Different controls operate at different layers.

- DHCP Snooping addresses rogue DHCP behavior.
- Port Security addresses unauthorized MAC addresses.
- BPDU Guard protects the spanning-tree boundary.
- ACLs control Layer 3 inter-VLAN communication.
- Management VLAN restrictions protect administrative access.
- The perimeter firewall provides the external security boundary.

This layered model prevents the security architecture from relying exclusively on the firewall.

6.9 Validation Status and Platform Limitation

The security design must be distinguished from the degree to which each control can be validated within Cisco Packet Tracer.

The ACL policies are defined as part of the intended Version 1 security architecture.

However, the validation environment imposes a limitation on Catalyst 3650/3560 ACL behavior: the intended `ip access-group` SVI enforcement does not reliably persist in the simulated running configuration.

Consequently, the report distinguishes between:

1. **policy design**, which is defined and traceable;
2. **configuration intent**, which is documented; and
3. **validated enforcement**, which is subject to the simulator limitation.

This distinction is important because successful validation of an ACL policy definition must not be represented as proof of full hardware-level ACL enforcement.

The limitation is therefore recorded as a Version 1 validation constraint rather than being hidden from the security assessment.

6.10 Security Design Baseline

The Version 1 security baseline consists of:

- VLAN and subnet-based security segmentation;
- least-privilege inter-VLAN ACL policy;
- guest isolation through RFC 1918 filtering;
- DHCP Snooping;
- Port Security;
- BPDU Guard and PortFast;
- dedicated Management VLAN 70;
- restricted administrative access;
- perimeter firewall protection; and
- explicit documentation of simulator-dependent enforcement limitations.

The security architecture therefore provides a layered control model while maintaining an explicit distinction between designed controls and controls that can be fully validated within the simulation environment.

This baseline is carried forward into Chapter 7 for implementation evidence and Chapter 8 for validation results.

Chapter 7 Implementation

This chapter documents the implementation of the Version 1 enterprise network baseline within the Cisco Packet Tracer validation environment.

The implementation translates the approved requirements, architecture, network design, and security controls into device-level configuration. The implementation is organized by infrastructure layer to maintain traceability between the engineering design and the resulting device configuration.

This chapter focuses on implementation evidence and configuration traceability. Detailed operational results are reserved for Chapter 8, while resilience and failure behavior are evaluated in Chapter 9.

7.1 Implementation Environment

The Version 1 implementation was developed and validated using Cisco Packet Tracer.

The simulated infrastructure represents a multi-floor enterprise campus network containing a centralized Layer 3 core, multiple access switches, perimeter security infrastructure, and an external routing boundary.

The principal infrastructure components are:

- CORE-L3-01 – primary Layer 3 core switch.
- CORE-L3-02 – secondary core switch and Layer 2 redundancy peer.
- ACC-F1-01 – Floor 1 access switch.
- ACC-F2-01 – Floor 2 access switch.
- ACC-F3-01 – Floor 3 access switch.
- FW-EDGE-01 – perimeter firewall.
- RTR-EDGE-01 – external/WAN routing boundary.

The implementation remains within the capabilities and limitations of the Packet Tracer environment. Platform-specific limitations are documented in Chapter 11.

7.2 Implementation Strategy

Implementation was performed in dependency order rather than configuring all devices simultaneously.

The implementation sequence was:

1. Establish device identity and baseline configuration.
2. Create the approved VLAN structure.
3. Configure trunk links and EtherChannels.
4. Configure spanning-tree behavior.
5. Configure Layer 3 SVIs and routing.
6. Apply access-layer security controls.
7. Apply Layer 3 access-control policies.
8. Configure management access.
9. Configure perimeter connectivity.
10. Perform implementation verification.
11. Record deviations and corrective changes.

This sequence provides a controlled dependency chain and makes it possible to identify implementation problems at the appropriate network layer.

7.3 Device Baseline

Each infrastructure device was assigned a deterministic hostname and configured according to its defined role.

The implementation roles are summarized below.

Table 7.1: Infrastructure Implementation Roles

Device	Implementation Role
CORE-L3-01	Primary L3 routing and policy enforcement
CORE-L3-02	Secondary core / L2 redundancy
ACC-F1-01	Floor 1 access switching
ACC-F2-01	Floor 2 access switching
ACC-F3-01	Floor 3 access switching
FW-EDGE-01	Perimeter firewall and NAT
RTR-EDGE-01	External/WAN routing

The resulting device configurations are maintained as project configuration artifacts rather than reproduced in full within this report.

7.4 VLAN Implementation

The approved VLAN structure was implemented across the relevant switching infrastructure.

The principal VLAN assignments are:

```

VLAN 10    DEV
VLAN 20    QA
VLAN 30    HR
VLAN 40    FIN
VLAN 50    EXEC
VLAN 60    SERVER
VLAN 70    MGMT/TRANSIT
VLAN 80    QUARANTINE
VLAN 90    GUEST
VLAN 999   NATIVE/DEAD

```

The VLAN configuration establishes the Layer 2 segmentation required by the network design.

7.4.1 VLAN Implementation Evidence

The implementation evidence is maintained in the corresponding access-switch and core-switch configuration files.

Table 7.2: VLAN Implementation Evidence

Implementation Area	Evidence Source	Verification Method
VLAN creation	Switch configuration	<code>show vlan brief</code>
VLAN membership	Access-switch configuration	<code>show vlan brief</code>
Trunk VLAN propagation	Trunk configuration	<code>show interfaces trunk</code>

The configuration files establish the intended implementation, while Chapter 8 records the observed operational state.

7.5 Trunk and EtherChannel Implementation

Infrastructure uplinks were implemented using 802.1Q trunking.

The core-to-access links were aggregated using LACP EtherChannels to provide logical link redundancy and increased aggregate capacity.

The implemented logical port-channels are:

```

Po1 -> ACC-F1-01
Po2 -> ACC-F2-01
Po3 -> ACC-F3-01

```

7.5.1 EtherChannel Implementation Evidence

The configuration evidence is distributed across the relevant core and access switch configuration files.

The actual operational state is evaluated during the validation and failure testing phases.

Table 7.3: EtherChannel Implementation Evidence

Evidence Item	Verification Method
LACP configuration	<code>show running-config</code>
Port-channel state	<code>show etherchannel summary</code>
Member interfaces	<code>show etherchannel summary</code>
Trunk state	<code>show interfaces trunk</code>

7.6 Spanning-Tree Implementation

Rapid-PVST+ was implemented as the Layer 2 spanning-tree protocol.

The intended root hierarchy is:

Primary Root:

CORE-L3-01

Secondary Root:

CORE-L3-02

This arrangement provides deterministic Layer 2 forwarding behavior while retaining an alternative core path for the Version 1 topology.

7.6.1 Spanning-Tree Evidence

Implementation evidence is maintained in the core and access-switch configuration files.

Table 7.4: Spanning-Tree Implementation Evidence

Evidence Item	Verification Method
Rapid-PVST+ configuration	<code>show running-config</code>
Root bridge state	<code>show spanning-tree</code>
Port forwarding state	<code>show spanning-tree</code>

Failure behavior and convergence are evaluated separately in Chapter 9.

7.7 Layer 3 Implementation

7.7.1 SVI Configuration

Layer 3 switching was implemented on CORE-L3-01 through Switched Virtual Interfaces.

The approved addressing plan follows the ‘/24’ subnet structure:

VLAN 10 -> 10.10.10.0/24

VLAN 20 -> 10.10.20.0/24

VLAN 30 -> 10.10.30.0/24

VLAN 40 -> 10.10.40.0/24

VLAN 50 -> 10.10.50.0/24
 VLAN 60 -> 10.10.60.0/24
 VLAN 70 -> 10.10.70.0/24
 VLAN 80 -> 10.10.80.0/24
 VLAN 90 -> 10.10.90.0/24

The SVI implementation provides the default gateway function for the corresponding routed VLANs.

7.7.2 Layer 3 Implementation Evidence

Table 7.5: Layer 3 Implementation Evidence

Implementation Area	Evidence Source	Verification Method
SVI configuration	CORE-L3-01.cfg	<code>show ip interface brief</code>
IP routing	CORE-L3-01.cfg	<code>show ip route</code>
Gateway reachability	Implementation topology	Connectivity test

This implementation provides the operational realization of the centralized Layer 3 architecture described in Chapters 4 and 5.

7.8 Access Layer Security Implementation

The access switches implement the endpoint-level security controls defined in Chapter 6.

The implementation includes, where applicable:

- Port Security.
- Sticky MAC learning.
- PortFast.
- BPDU Guard.
- DHCP Snooping.

These controls are configured on appropriate endpoint-facing interfaces and infrastructure paths.

7.8.1 Security Implementation Evidence

The detailed security rationale is documented in Chapter 6, while operational effectiveness is evaluated in Chapter 8.

7.9 Extended ACL Implementation

Extended ACLs were implemented on CORE-L3-01 to enforce the approved inter-VLAN security policies.

Table 7.6: Access Layer Security Implementation Evidence

Control	Verification Method
Port Security	<code>show port-security</code>
Sticky MAC	<code>show port-security address</code>
PortFast / BPDU Guard	<code>show spanning-tree</code>
DHCP Snooping	<code>show ip dhcp snooping</code>
DHCP Bindings	<code>show ip dhcp snooping binding</code>

The implementation provides the Layer 3 enforcement point for:

- Departmental access restrictions.
- Sensitive-zone protection.
- Server access control.
- Guest network isolation.
- Management-plane restrictions.

7.9.1 ACL Implementation Evidence

Table 7.7: ACL Implementation Evidence

Evidence Item	Verification Method
ACL definitions	<code>show access-lists</code>
ACL interface binding	<code>show ip interface</code>
Routing state	<code>show ip route</code>
Policy behavior	Chapter 8 traffic tests

The configuration source establishes the intended policy, while validation tests establish whether the policy produces the expected traffic behavior.

7.10 Management Implementation

The infrastructure management boundary was implemented through VLAN 70.

The documented management protocols are:

- SSH version 2 for remote administration.
- SNMP version 3 for infrastructure monitoring.

Management access is separated from ordinary endpoint traffic and is subject to the corresponding Layer 3 security policy.

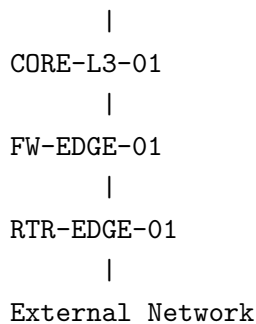
Table 7.8: Management Implementation Evidence

Implementation Area	Verification Method
Management SVI	<code>show ip interface brief</code>
SSH configuration	<code>show running-config</code>
SNMP configuration	<code>show running-config</code>
Management connectivity	Chapter 8 validation

7.11 Perimeter Implementation

The external connectivity path was implemented according to the approved architecture:

Internal Network



FW-EDGE-01 provides the documented perimeter and NAT function, while RTR-EDGE-01 represents the external/WAN routing boundary.

The implementation evidence is maintained in the corresponding device configuration artifacts.

Table 7.9: Perimeter Implementation Evidence

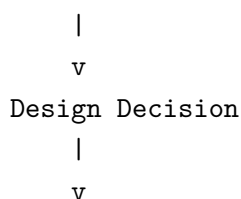
Implementation Area	Verification Method
Firewall configuration	Device configuration review
NAT configuration	Firewall operational verification
External routing	Routing-state verification
End-to-end connectivity	Chapter 8 validation

7.12 Implementation Evidence Register

The project uses configuration artifacts as the primary implementation evidence.

The evidence chain follows:

Requirement



Device Configuration

|

v

Operational Verification

|

v

Validation Result

The following register summarizes the principal implementation evidence.

Table 7.10: Implementation Evidence Register

Area	Device	Evidence Source	Verification
VLAN/SVI	CORE-L3-01	CORE-L3-01.cfg	show ip int brief
VLAN Access	Access Switches	Device configuration	show vlan brief
EtherChannel	Core/Access	Device configuration	show etherchannel summary
Rapid-PVST+	Core/Access	Device configuration	show spanning-tree
ACL	CORE-L3-01	CORE-L3-01.cfg	show access-lists
Port Security	Access Switches	Device configuration	show port-security
DHCP Snooping	Access Switches	Device configuration	show ip dhcp snooping
Management	Core/Infrastructure	Device configuration	SSH/SNMP verification
Perimeter/NAT	FW-EDGE-01	Firewall configuration	Operational verification

The register provides a compact representation of the relationship between implemented functionality and its corresponding evidence source.

7.13 Implementation Deviation and Change Record

During implementation, a physical connectivity deviation involving the development endpoint was identified.

The affected endpoint was:

DEV-PC-01

VLAN 10

ACC-F2-01:Fa0/1

The deviation was not treated as an undocumented correction. Instead, it was processed through the project's change-management workflow.

The corresponding change record is:

CHANGE-001

The change record documents the affected scope, corrective action, verification activity, and resulting documentation update.

This demonstrates that the implementation process includes configuration governance and traceability in addition to technical deployment.

7.14 Implementation Baseline

The Version 1 implementation establishes the following engineering baseline:

1. The approved VLAN and IPv4 addressing structure was implemented.
2. Core SVI routing was implemented according to the centralized Layer 3 design.
3. Core-to-access uplinks were implemented using LACP EtherChannels.
4. Rapid-PVST+ was implemented with the defined core root hierarchy.
5. Access-layer security controls were implemented on appropriate endpoint-facing interfaces.
6. Extended ACLs were implemented at the Layer 3 enforcement point.
7. Management access was separated through VLAN 70.
8. The documented perimeter connectivity model was implemented.
9. Implementation deviations were recorded through formal change management.

The implementation therefore represents the operational realization of the architecture, network design, and security baseline defined in Chapters 4–6.

Chapter 8 evaluates whether the resulting implementation satisfies the defined functional and security requirements through structured validation tests.

Chapter 8 Validation

This chapter presents the formal validation of the Version 1 Enterprise Network Infrastructure Blueprint.

Validation was performed against the implemented Cisco Packet Tracer baseline using defined acceptance test cases. The purpose of this phase was to determine whether the implemented infrastructure satisfied the approved functional, security, and resilience requirements.

The validation process deliberately distinguishes between intended configuration and observed behavior. Configuration evidence establishes what was implemented, while validation tests establish whether the implementation behaved as expected.

8.1 Validation Methodology

The Version 1 validation cycle was organized into twelve engineering phases covering Layer 2 infrastructure, Layer 3 routing, security controls, availability, failure behavior, and implementation governance.

A formal acceptance-test approach was used. Each test case defines:

- the target component or network function;
- the required preconditions;
- the verification or test procedure;
- the expected behavior;
- the observed validation result; and
- the resulting PASS, PARTIAL, or accepted limitation status.

The authoritative acceptance-test definitions are maintained in the project validation artifacts.

The validation chain follows:

Requirement

|

v

Implementation

|

v

Test Case

|

v
Observed Result
|
v
Acceptance Status

This approach maintains traceability between the engineering requirements, implemented controls, and final acceptance decision.

8.2 Validation Environment

Validation was conducted within the Cisco Packet Tracer implementation environment defined by the Version 1 engineering baseline.

The principal validation targets included:

- CORE-L3-01
- CORE-L3-02
- ACC-F1-01
- ACC-F2-01
- ACC-F3-01
- FW-EDGE-01
- representative endpoint and server nodes

The validation environment is a simulated engineering environment rather than a production deployment. Consequently, observed results are interpreted within the documented capabilities and limitations of Cisco Packet Tracer.

8.3 Validation Scope

The formal validation scope covers four principal engineering domains:

1. Layer 2 switching and physical topology behavior.
2. Layer 3 routing and connectivity.
3. Security and access-control enforcement.
4. High availability and failure behavior.

The formal acceptance-test specification contains 15 test cases executed through the Phase 1–12 validation cycle.

The test cases are grouped into the following validation domains.

Table 8.1: Validation Test Domains

Domain	Test Group	Primary Objective
Layer 2	TC-L2	Switching and topology integrity
Layer 3	TC-L3	Routing and gateway connectivity
Security	TC-SEC	Access-control enforcement
High Availability	TC-HA	Failure and resilience behavior

8.4 Layer 2 Validation

Layer 2 validation verifies VLAN segmentation, trunking, link aggregation, spanning-tree behavior, and access-edge protection.

8.4.1 TC-L2-01: VLAN Trunking and 802.1Q Encapsulation

The first Layer 2 test verifies the Core-to-Access trunk configuration.

The test target consists of the interconnection between CORE-L3-01 and the access switches.

The verification procedure uses:

```
show interfaces trunk
```

The expected state includes:

- 802.1Q trunk encapsulation;
- approved VLANs 10–90 and VLAN 999;
- native VLAN 999;
- active VLAN propagation across the trunk.

The test verifies that the Layer 2 transport required by the architecture is operational before higher-layer routing and security validation are performed.

8.4.2 TC-L2-02: LACP EtherChannel

LACP implementation was validated on the access uplink represented by Port-Channel 1.

The verification procedure uses:

```
show etherchannel summary
```

The expected state is an operational LACP bundle with the logical Port-Channel in service and member interfaces participating in the bundle.

The test establishes the operational baseline required for subsequent link-failure testing.

8.4.3 TC-L2-03: Rapid-PVST+ Root Bridge Election

Rapid-PVST+ root election was validated across the switching topology.

The verification procedure uses:

```
show spanning-tree vlan 10
```

The intended root hierarchy places CORE-L3-01 as the primary root, with CORE-L3-02 providing the secondary Layer 2 role.

The expected topology includes the alternate path on CORE-L3-02 remaining blocked until the primary path becomes unavailable.

8.4.4 TC-L2-04: Edge Protection

The access-layer edge protection baseline was validated for PortFast and BPDU Guard.

The test target covers endpoint-facing access interfaces.

The expected security behavior is that an unauthorized BPDU received on an edge-protected interface causes the interface to enter the appropriate protective state rather than allowing a Layer 2 loop to propagate.

This test validates the Layer 2 edge-defense control defined in Chapter 6.

8.5 Layer 3 Validation

Layer 3 validation verifies gateway reachability, inter-VLAN routing, and default egress routing.

8.5.1 TC-L3-01: SVI Gateway Reachability

Core SVI gateways were validated across the defined routed VLANs.

The target gateway range includes:

```
10.10.10.1  
10.10.20.1  
10.10.30.1  
10.10.40.1  
10.10.50.1  
10.10.60.1  
10.10.70.1  
10.10.80.1  
10.10.90.1
```

The test procedure uses endpoint-to-gateway ICMP connectivity.

The acceptance criterion is successful gateway reachability from the corresponding VLAN.

8.5.2 TC-L3-02: Inter-VLAN Routing

Inter-VLAN routing was validated using traffic from the development zone to the server zone.

The representative test path is:

```
DEV-PC-01
10.10.10.10
  |
  v
CORE-L3-01
  |
  v
Git Server
10.10.60.21
```

The test verifies that Layer 3 forwarding operates between authorized network segments according to the defined routing and security policy.

8.5.3 TC-L3-03: Default Egress Routing

Default internal egress routing was validated toward the perimeter firewall.

The representative gateway is:

```
FW-EDGE-01
10.10.70.2
```

The validation confirms that internal traffic can reach the firewall through the designated management/transit segment and the corresponding Layer 3 routing path.

8.6 Security Validation

Security validation evaluates whether the implemented access-control mechanisms enforce the intended segmentation and isolation policies.

8.6.1 TC-SEC-01: Extended ACL Traffic Filtering

The development-zone ACL was tested against access to the protected server zone.

The representative traffic path is:

```
DEV-PC-01
10.10.10.10
  |
  x
```

|
Git Server
10.10.60.21

The expected policy result is that unauthorized ICMP traffic is blocked.

The validation confirmed the intended policy behavior.

However, the validation record also documents a Cisco Packet Tracer limitation involving the handling/parsing of Extended ACL behavior on SVI interfaces.

This limitation is treated as a documented environment constraint rather than being presented as evidence of unrestricted production behavior.

8.6.2 TC-SEC-02: Guest Zone Isolation

Guest isolation was validated from VLAN 90 toward protected internal resources.

The representative test path is:

Guest PC
10.10.90.10
|
X
|
Internal Server
10.10.60.21

The expected result is that guest traffic cannot establish unauthorized lateral access to internal subnets.

This test validates the guest-isolation requirement established in the requirements and security design chapters.

8.7 High Availability and Chaos Validation

Availability validation intentionally introduces controlled failures into the network to evaluate the behavior of redundant Layer 2 paths and aggregated links.

These tests are distinguished from ordinary functional validation because they intentionally modify the operating state of the topology.

8.7.1 TC-HA-01: Core Trunk Link Failure

The primary Core trunk path was intentionally disabled.

The failure condition was introduced using:

shutdown

on the designated primary trunk interface.

The expected behavior is for the alternate Layer 2 path through **CORE-L3-02** to transition from its alternate blocking state into the forwarding state following Rapid-PVST+ convergence.

This test validates the Layer 2 resilience mechanism without claiming full Layer 3 gateway redundancy.

8.7.2 TC-HA-02: LACP Member Link Failure

A member link of the access-layer EtherChannel was intentionally disabled.

The expected behavior is that the logical Port-Channel remains operational through the surviving member.

The validation verifies the resilience of the aggregated Layer 2 path rather than relying on a single physical interface.

The test therefore establishes that a single member-link failure does not necessarily cause the logical uplink to become unavailable.

8.7.3 TC-HA-03: Active Core Layer 3 Gateway Failure

The active Layer 3 core was intentionally isolated to evaluate the effect of losing the primary gateway.

The expected and observed behavior is treated as an explicit Version 1 architectural limitation.

CORE-L3-01 provides the active Layer 3 gateway function, while **CORE-L3-02** operates as the secondary Layer 2 bridge.

Consequently, loss of **CORE-L3-01** results in interruption of Layer 3 routing.

This condition is ****not** classified as an implementation defect** because full first-hop Layer 3 redundancy was intentionally deferred from Version 1.

The corresponding future implementation is HSRP/VRRP-based gateway redundancy in Version 2.

8.8 Validation Results

The formal validation cycle achieved an overall acceptance result of approximately 94.4 percent PASS.

The result demonstrates that the majority of the implemented functional, security, and resilience controls satisfied their defined acceptance criteria.

The validation result should therefore be interpreted as an engineering acceptance baseline rather than a claim of unrestricted production readiness.

Table 8.2: Version 1 Validation Summary

Validation Area	Result
Layer 2 infrastructure	PASS
Layer 3 routing and connectivity	PASS
Access-layer security controls	PASS
Extended ACL policy validation	PASS / documented limitation
LACP resilience	PASS
Rapid-PVST+ failure convergence	PASS
Layer 3 gateway redundancy	PARTIAL
Overall Version 1	94.4% PASS

8.9 Validation Evidence

The project maintains validation evidence through test-case definitions, device configuration artifacts, operational verification commands, and documented results.

Representative verification commands include:

```
show interfaces trunk
show etherchannel summary
show spanning-tree
show ip interface brief
show ip route
show access-lists
show port-security
show ip dhcp snooping
```

These commands are used as evidence-generation mechanisms rather than as substitutes for the formal acceptance criteria.

The complete command output is intentionally not reproduced throughout this report. This keeps the final engineering document focused on results and traceability while retaining the underlying configuration and validation artifacts in the project repository.

8.10 Requirement-to-Test Traceability

The validation process maintains a direct relationship between requirements and their corresponding test cases.

The complete requirement mapping remains maintained in the project requirements and validation artifacts.

8.11 Known Validation Limitations

Two limitations are particularly important when interpreting the Version 1 results.

First, Cisco Packet Tracer does not reproduce every behavior of a physical Cisco production

Table 8.3: Requirement-to-Validation Traceability

Requirement Area	Validation Coverage
VLAN segmentation	TC-L2-01, TC-L2-04
DHCP and endpoint services	Functional validation set
Inter-VLAN routing	TC-L3-01, TC-L3-02
Default egress	TC-L3-03
Guest isolation	TC-SEC-02
Extended ACL protection	TC-SEC-01
LACP availability	TC-L2-02, TC-HA-02
Rapid-PVST+ resilience	TC-L2-03, TC-HA-01
Access-layer protection	TC-L2-04
Core availability limitation	TC-HA-03

environment. In particular, the validation record identifies limitations affecting Extended ACL behavior and parser support.

Second, Version 1 intentionally implements an active-passive Layer 2 core redundancy model rather than full first-hop Layer 3 gateway redundancy.

Therefore, the loss of CORE-L3-01 interrupts centralized Layer 3 routing.

This limitation is formally accepted within the Version 1 scope and is not hidden from the validation result.

8.12 Validation Acceptance

The Version 1 baseline is considered accepted within its defined engineering scope.

The acceptance decision is based on:

- completion of the formal validation cycle;
- successful validation of the principal Layer 2 and Layer 3 functions;
- successful validation of the defined security controls;
- successful LACP and Rapid-PVST+ resilience testing;
- explicit documentation of Packet Tracer limitations;
- explicit acceptance of the Version 1 Layer 3 gateway redundancy limitation.

The resulting status is:

VERSION 1 – ACCEPTED & FROZEN

The baseline therefore represents a completed portfolio engineering implementation rather than an unfinished prototype.

Further architectural improvements, including first-hop Layer 3 gateway redundancy, are treated as Version 2 evolution and are documented separately from the accepted Version 1 baseline.

8.13 Validation Conclusion

The validation phase demonstrates that the Version 1 implementation substantially satisfies the engineering requirements defined for the Enterprise Network Infrastructure Blueprint.

The achieved 94.4 percent PASS result, combined with explicit documentation of the remaining limitations, provides an auditable engineering basis for the Version 1 portfolio baseline.

The validation process also establishes the evidence required for the next phase of the report: detailed failure and resilience analysis in Chapter 9.

Chapter 9 Failure and Resilience

This chapter evaluates the behavior of the Version 1 network infrastructure under controlled failure conditions.

Unlike functional validation in Chapter 8, which verifies whether individual requirements operate correctly under normal conditions, this chapter intentionally introduces faults into the topology and observes the resulting network behavior.

The objective is not to demonstrate theoretical fault tolerance. Instead, the objective is to establish an evidence-based understanding of which failure modes are absorbed by the Version 1 architecture, which require convergence, and which remain accepted architectural limitations.

9.1 Failure Testing Philosophy

Failure testing was performed using controlled fault injection.

The general procedure was:

1. Establish the normal operating state.
2. Identify the redundant or failure-sensitive path.
3. Introduce a controlled failure.
4. Observe the resulting topology state.
5. Verify whether an alternate path becomes operational.
6. Restore the failed component.
7. Confirm recovery to the intended baseline.

This approach allows resilience behavior to be evaluated without confusing normal functional correctness with failure recovery.

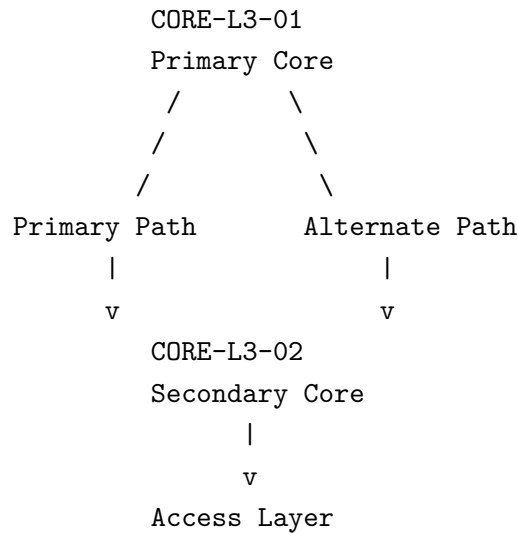
The principal failure domains tested in Version 1 are:

- Core-to-Core Layer 2 path failure.
- EtherChannel member-link failure.
- Active Layer 3 core failure.

9.2 Resilience Model

The Version 1 topology provides redundancy at selected Layer 2 infrastructure points.

The simplified resilience model is:



At the access-uplink level, LACP provides logical redundancy by aggregating multiple physical interfaces into a single Port-Channel.

At the spanning-tree level, Rapid-PVST+ maintains an alternate path that can be activated when the preferred Layer 2 path fails.

However, the Version 1 design does not provide first-hop Layer 3 gateway redundancy. CORE-L3-01 remains the active Layer 3 gateway.

This distinction is central to interpreting the failure results.

9.3 TC-HA-01: Core-to-Core Trunk Link Failure

9.3.1 Failure Scenario

The first resilience test evaluates the failure of the primary Core-to-Core Layer 2 trunk.

The affected interface is:

```

CORE-L3-01
Gi1/0/23
  
```

The failure is introduced by administratively shutting down the interface:

```
shutdown
```

The purpose of this test is to determine whether the alternate Layer 2 path through CORE-L3-02 becomes available after Rapid-PVST+ convergence.

9.3.2 Expected Behavior

Before the failure, the alternate path is maintained in a blocking state by Rapid-PVST+ in order to prevent a Layer 2 loop.

The expected state transition is:

Normal State

Primary Path

|

v

Forwarding

Alternate Path

|

v

Altn / Blocking

After the primary path fails:

Primary Path

|

X

|

FAILED

Alternate Path

|

v

Root / Forwarding

The acceptance criterion is that the alternate path transitions into the forwarding state after RSTP convergence.

9.3.3 Engineering Significance

This test demonstrates that the Version 1 topology does not rely exclusively on a single Layer 2 physical path.

The redundant path remains intentionally blocked during normal operation and becomes available only when the primary forwarding path is lost.

This provides deterministic loop prevention while retaining a controlled failure-recovery mechanism.

9.4 TC-HA-02: LACP Member-Link Failure

9.4.1 Failure Scenario

The second resilience test evaluates the failure of one physical member of the access-layer EtherChannel.

The affected interface is:

ACC-F1-01
Fa0/23

The interface is administratively disabled while the second EtherChannel member remains operational.

The logical bundle is:

```
Fa0/23 ----\
              \
              > Po1
              /
Fa0/24 ----/
```

9.4.2 Expected Behavior

The expected behavior is that the logical Port-Channel remains operational after the loss of one member.

The remaining interface should continue carrying traffic:

```
Fa0/23
  X
  |
  FAILED

Fa0/24
  |
  v
Po1
  |
  v
Traffic continues
```

The formal acceptance criterion is that Po1 remains operational and the surviving member continues forwarding traffic.

9.4.3 Resilience Interpretation

This test demonstrates the difference between physical-link redundancy and logical-link availability.

A single physical interface failure does not necessarily remove the logical uplink because LACP maintains the aggregated Port-Channel using the surviving member.

This is an important resilience property for the access-layer infrastructure because it reduces dependence on any individual physical uplink.

9.5 TC-HA-03: Active Core Layer 3 Gateway Failure

9.5.1 Failure Scenario

The third test intentionally isolates the active Layer 3 core:

CORE-L3-01

Active L3 Gateway

The test isolates the interfaces associated with the active Core and observes the resulting Layer 3 behavior.

Unlike the previous two tests, this test is expected to expose a deliberate Version 1 architectural limitation.

9.5.2 Observed Architecture

The Version 1 design separates Layer 2 redundancy from Layer 3 gateway redundancy.

The resulting model is:

CORE-L3-01

|

+++ Active L3 SVI Gateways

|

+++ L3 Routing

|

X

|

Failure

|

v

CORE-L3-02

|

+++ Secondary L2 role

|

+++ No equivalent active L3 gateway

Consequently, the secondary Core can continue to participate in the Layer 2 topology but cannot transparently replace the failed Layer 3 gateway.

9.5.3 Failure Result

When CORE-L3-01 becomes unavailable, Layer 3 traffic is interrupted.

This behavior is classified as:

PARTIAL – ACCEPTED VERSION 1 LIMITATION

It is not classified as an unexpected implementation failure.

The limitation is consistent with the approved Version 1 scope, which uses an active-passive Layer 2 redundancy model and deliberately defers first-hop gateway redundancy.

9.6 Failure Behavior Summary

The three formal failure scenarios produce different resilience outcomes.

Table 9.1: Version 1 Failure and Resilience Results

Test	Failure	Result
TC-HA-01	Core trunk failure	PASS – alternate STP path activates
TC-HA-02	EtherChannel member failure	PASS – surviving member maintains Po1
TC-HA-03	Active L3 Core failure	PARTIAL – accepted gateway limitation

The results demonstrate that Version 1 provides meaningful resilience at the Layer 2 infrastructure level while retaining an explicit Layer 3 single gateway limitation.

9.7 Failure Domain Analysis

The failure tests can be interpreted according to the layer at which redundancy is implemented.

Table 9.2: Failure Domain Analysis

Failure Domain	Mechanism	V1 Behavior
Core trunk	Rapid-PVST+	Alternate path available
Physical uplink	LACP	Surviving member continues forwarding
Layer 3 gateway	No FHRP in V1	Service interruption

This analysis demonstrates an important architectural boundary.

The project does not claim that “redundant Core” automatically means “fully redundant network.”

Instead, redundancy is evaluated at the individual control and failure domain level.

9.8 Recovery Considerations

The failure tests also establish the expected recovery mechanisms.

For Layer 2 failures, Rapid-PVST+ and LACP provide mechanisms for restoring forwarding through surviving paths.

For an EtherChannel member failure, recovery occurs through continued operation of the logical Port-Channel using the remaining member.

For an active Layer 3 gateway failure, no equivalent automatic first-hop gateway transition exists in Version 1.

Therefore, Layer 3 recovery requires restoration of the active Core or implementation of a future first-hop redundancy mechanism.

9.9 Engineering Trade-off

The Version 1 architecture deliberately prioritizes a controlled and traceable implementation scope over maximum redundancy.

Introducing HSRP or VRRP would remove the identified Layer 3 single-gateway limitation, but it would also expand the Version 1 implementation and validation scope.

The project therefore treats first-hop redundancy as a Version 2 architectural evolution rather than introducing it solely to improve the Version 1 validation score.

This decision preserves the integrity of the frozen engineering baseline.

9.10 Resilience Limitations

The resilience conclusions must be interpreted within the Cisco Packet Tracer environment.

Packet Tracer provides a useful controlled simulation environment for demonstrating switching, routing, spanning-tree, EtherChannel, and security behavior, but it does not reproduce every characteristic of physical production infrastructure.

Accordingly, the failure tests demonstrate architectural behavior within the simulation environment rather than guaranteeing production-level recovery times or availability figures.

No production SLA is inferred from the observed convergence behavior.

9.11 Failure Testing Conclusion

The Version 1 failure tests demonstrate that the architecture provides controlled Layer 2 resilience through Rapid-PVST+ and LACP.

The Core trunk failure test demonstrates alternate-path activation, while the EtherChannel test demonstrates continued logical-uplink operation after a single member failure.

The active Core failure test confirms the known Layer 3 gateway limitation.

The resulting resilience profile can therefore be summarized as:

Layer 2 Resilience: VALIDATED

Layer 3 Gateway Redundancy: DEFERRED TO V2

This outcome is consistent with the accepted Version 1 architecture and provides a clear engineering basis for the future evolution described in Chapter 14.

Chapter 10 Change Management

This chapter documents the engineering change-management process applied to the Version 1 network baseline.

The purpose of change management is to ensure that modifications to the network are introduced in a controlled, traceable, and reversible manner.

Rather than treating configuration changes as isolated CLI operations, the project treats each significant change as an engineering event with a defined scope, implementation target, validation procedure, and rollback consideration.

10.1 Change Management Principles

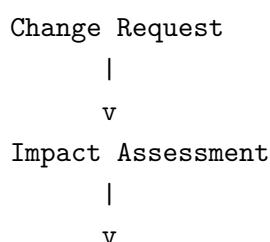
The Version 1 change-management process follows five principles:

1. **Traceability** – every significant change must be associated with a defined requirement, scenario, or engineering objective.
2. **Controlled Scope** – a change should affect only the components required to implement the intended outcome.
3. **Validation** – the resulting state must be verified against the expected behavior.
4. **Reversibility** – changes should have a defined rollback or restoration path where technically applicable.
5. **Evidence Preservation** – the final configuration and validation result must remain traceable to the change record.

These principles prevent uncontrolled configuration drift and establish a repeatable workflow for future revisions of the network.

10.2 Change Lifecycle

The project uses the following simplified change lifecycle:



Implementation Planning

|

v

Controlled Change

|

v

Validation

|

v

Evidence Capture

|

v

Change Closure

A change is not considered complete merely because the requested configuration has been entered.

Closure requires confirmation that the resulting network state satisfies the intended objective and does not introduce an unaccepted regression.

10.3 CHANGE-001: Development Endpoint Placement

The primary documented change scenario for the Version 1 baseline is **CHANGE-001**.

The change concerns the placement of the development endpoint **DEV-PC-01** within the enterprise access network.

The endpoint is assigned to:

Device: DEV-PC-01
VLAN: 10
Access Switch: ACC-F2-01
Interface: Fa0/1

The change therefore provides a concrete example of how an endpoint-level requirement is translated into an implementation-level network change.

10.4 Change Objective

The objective of **CHANGE-001** is to place the development endpoint in the designated Development VLAN while maintaining the intended access-layer security and network segmentation controls.

The target state is:

DEV-PC-01


```
    |
    v
ACC-F2-01
Fa0/1
    |
    v
VLAN 10
Development Zone
    |
    v
CORE-L3-01
```

The change is intentionally limited to the endpoint's access-layer assignment.

No unrelated Core routing, firewall, or security-policy modification is required to implement this change.

10.5 Impact Assessment

Before implementation, the change is evaluated against the surrounding network dependencies.

The primary impact areas are:

- endpoint VLAN membership;
- access-port configuration;
- DHCP address allocation;
- Layer 3 gateway reachability;
- ACL policy applied to the Development zone;
- connectivity to authorized internal services.

The change does not require modification of the Core SVI gateway or the existing inter-VLAN routing architecture.

The expected impact is therefore localized to the access edge and the endpoint's resulting Layer 2 and Layer 3 membership.

10.6 Implementation Procedure

The implementation sequence follows a controlled configuration process.

10.6.1 Step 1: Identify the Target

The target endpoint and access interface are identified before configuration:

```
DEV-PC-01
      |
      v
ACC-F2-01:Fa0/1
```

This prevents accidental modification of another endpoint or access port.

10.6.2 Step 2: Assign the Access VLAN

The access interface is configured for the Development VLAN:

```
interface Fa0/1
  switchport mode access
  switchport access vlan 10
```

The interface therefore becomes a member of VLAN 10.

10.6.3 Step 3: Verify the Local Configuration

The resulting configuration is verified using the switch configuration and VLAN state.

Representative commands include:

```
show vlan brief
show running-config interface Fa0/1
```

The expected result is that Fa0/1 is associated with VLAN 10.

10.6.4 Step 4: Verify Endpoint Connectivity

After the VLAN assignment is confirmed, endpoint connectivity is validated.

The verification sequence includes:

```
DEV-PC-01
      |
      v
10.10.10.1
Development Gateway
      |
      v
Authorized Network Services
```

The purpose is to confirm that the endpoint receives the correct network membership and can reach resources permitted by the Development-zone policy.

10.7 Change Validation

Change validation evaluates both the intended positive behavior and the absence of unintended side effects.

The validation criteria include:

Table 10.1: CHANGE-001 Validation Criteria

Validation Item	Expected Result
Access-port assignment	VLAN 10
Endpoint VLAN membership	Development zone
Default gateway	Development SVI
DHCP behavior	Address from VLAN 10 scope
Authorized connectivity	Permitted by policy
Unauthorized connectivity	Blocked by policy
Unrelated infrastructure	No unintended modification

This validation approach ensures that the change is evaluated as a complete network behavior rather than merely as a successful configuration command.

10.8 Configuration Evidence

The change record is supported by configuration evidence from the affected access switch.

The evidence chain is:

CHANGE-001

|

v

ACC-F2-01

|

v

Fa0/1

|

v

VLAN 10

|

v

DEV-PC-01

The configuration artifact establishes the intended implementation state, while the validation artifact establishes the resulting operational state.

This distinction is important because a configuration file alone does not prove that the desired network behavior was achieved.

10.9 Rollback Strategy

The change is designed to be reversible.

If the endpoint must be removed from VLAN 10, the access-port assignment can be restored to its previous approved state.

The rollback procedure follows the same controlled sequence:

1. Identify the affected interface.
2. Restore the previous approved access configuration.
3. Verify VLAN membership.
4. Verify endpoint connectivity.
5. Confirm that unrelated infrastructure remains unchanged.

The rollback concept therefore preserves the same traceability requirements as the original change.

10.10 Change Risk Assessment

The primary risks associated with CHANGE-001 are localized to the access edge.

Table 10.2: CHANGE-001 Risk Assessment

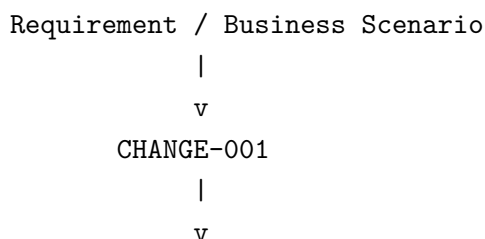
Risk	Impact	Control
Wrong VLAN assignment	Endpoint isolation / incorrect policy	Post-change VLAN verification
Wrong access interface	Incorrect endpoint placement	Explicit device/interface identification
Unexpected connectivity	Security-policy violation	Post-change connectivity testing
Configuration regression	Service interruption	Configuration verification and rollback

The change is therefore considered a localized access-layer modification rather than a Core-level architectural change.

10.11 Change Traceability

The change-management process maintains bidirectional traceability.

The forward relationship is:



```

ACC-F2-01:Fa0/1
  |
  v
  VLAN 10
  |
  v
DEV-PC-01

```

The reverse relationship is equally important.

Given the observed configuration of DEV-PC-01, the engineering team can identify the corresponding switch interface, VLAN assignment, and original change scenario.

This provides a practical mechanism for configuration accountability.

10.12 Baseline Protection

After validation, the accepted configuration becomes part of the Version 1 engineering baseline.

Subsequent modifications should not silently overwrite the accepted state.

Instead, future changes should:

- reference the affected baseline component;
- identify the reason for modification;
- document the expected impact;
- undergo appropriate validation; and
- update the engineering evidence when the change is accepted.

This process helps prevent configuration drift between the documented architecture and the actual implemented state.

10.13 Version 1 Change Governance

The Version 1 change-management approach is intentionally lightweight.

The project is a portfolio engineering implementation rather than a production enterprise operated under a formal ITIL change advisory board.

Nevertheless, the engineering principles are preserved:

```

Controlled Scope
  +
Traceability
  +

```

Validation

+

Rollback

+

Evidence

=

Controlled Change

This provides a practical bridge between academic network implementation and professional infrastructure engineering practice.

10.14 Change Management Conclusion

CHANGE-001 demonstrates that the Version 1 network is not treated as a collection of isolated device configurations.

The endpoint placement scenario is represented as a traceable engineering change from the original requirement through implementation and validation.

The process establishes:

- controlled modification of the network baseline;
- explicit identification of the affected device and interface;
- validation of the resulting network state;
- a defined rollback concept; and
- preservation of configuration evidence.

The resulting change-management process provides the governance foundation required to maintain consistency between the approved engineering design, the implemented configuration, and subsequent revisions of the network.

Chapter 11 Limitations

This chapter documents the known limitations of the Version 1 Enterprise Network Infrastructure Blueprint.

The purpose of this chapter is not to weaken the engineering baseline, but to establish a clear boundary between validated Version 1 capabilities, accepted architectural constraints, and improvements intentionally deferred to future revisions.

A limitation is therefore not automatically classified as an implementation defect. Each limitation is evaluated according to its origin, engineering impact, and relationship to the approved Version 1 scope.

11.1 Limitation Classification

The limitations identified in this project are divided into three categories:

1. **Architectural limitations** – constraints resulting from deliberate Version 1 design decisions.
2. **Simulation limitations** – constraints introduced by the Cisco Packet Tracer validation environment.
3. **Scope limitations** – capabilities intentionally excluded from the Version 1 implementation.

This classification prevents simulation behavior from being incorrectly interpreted as a production-network limitation and prevents intentionally deferred functionality from being classified as an implementation failure.

11.2 Version 1 Layer 3 Gateway Limitation

The most significant architectural limitation is the absence of first-hop Layer 3 gateway redundancy.

The Version 1 Core architecture provides redundancy at the Layer 2 level, while CORE-L3-01 remains the active Layer 3 gateway.

The simplified model is:

```
CORE-L3-01
Active L3
|
```

SVI / Routing
|
X
Failure
|
v
CORE-L3-02
Secondary L2

If CORE-L3-01 becomes unavailable, CORE-L3-02 does not automatically assume the Layer 3 gateway role.

Consequently, Layer 3 connectivity is interrupted until the active gateway is restored.

This condition was explicitly tested through TC-HA-03 and is accepted as a Version 1 architectural limitation.

11.2.1 Reason for Deferral

First-hop redundancy using a protocol such as HSRP or VRRP was intentionally deferred to a future version.

The decision preserves the scope of the Version 1 implementation and avoids introducing an additional high-availability mechanism without completing its associated design, implementation, and validation cycle.

The resulting engineering decision is:

V1: Layer 2 Resilience
V2: First-Hop Layer 3 Redundancy

This distinction is important because the project does not claim full end-to-end redundancy where none has been implemented.

11.3 Cisco Packet Tracer Simulation Limitations

The implementation and validation environment is Cisco Packet Tracer.

Packet Tracer provides an effective environment for demonstrating the majority of the project's switching, routing, segmentation, security, and resilience concepts.

However, it is a simulation environment and does not reproduce every behavior of physical Cisco infrastructure.

The validation results must therefore be interpreted as simulation-based engineering evidence rather than as production certification.

11.3.1 Extended ACL Behavior

The validation record identifies limitations associated with Extended ACL behavior and parser support within the Packet Tracer environment.

The project therefore distinguishes between:

Intended Security Policy

|

v

Cisco Configuration

|

v

Packet Tracer Behavior

|

v

Observed Limitation

An observed simulator limitation is not automatically treated as evidence that the underlying production Cisco architecture would behave identically.

The limitation is retained in the validation record so that the final engineering baseline remains transparent.

11.4 Absence of Physical Production Validation

The Version 1 implementation was not deployed on physical Cisco hardware.

Consequently, the project does not claim empirical validation of:

- physical interface behavior;
- hardware failure characteristics;
- real switch boot and recovery behavior;
- physical transceiver or cabling faults;
- production-grade convergence measurements;
- hardware-specific performance characteristics.

The validation evidence therefore demonstrates logical and simulated behavior within the defined environment.

Production deployment would require an additional hardware validation phase.

11.5 Performance and Capacity Limitations

Version 1 focuses on architecture, segmentation, security controls, resilience, and functional validation.

It does not establish production capacity benchmarks.

The project does not claim validated values for:

- maximum throughput;
- packets-per-second capacity;
- concurrent-user capacity;
- application response time;
- firewall throughput under production load;
- long-duration utilization;
- production SLA availability.

These measurements require physical or virtual infrastructure capable of generating representative production workloads.

Accordingly, no production performance target should be inferred from the Packet Tracer validation results.

11.6 Monitoring and Operational Visibility

The Version 1 implementation focuses primarily on infrastructure design and functional validation.

A complete production observability stack is outside the current scope.

The project does not implement a dedicated:

- centralized SIEM platform;
- network-management platform;
- centralized syslog infrastructure;
- NetFlow/IPFIX analytics platform;
- enterprise alerting pipeline;
- long-term configuration-compliance platform.

The absence of these systems does not invalidate the implemented security controls, but it limits the operational monitoring capabilities of the baseline.

11.7 Configuration Management Limitations

The Version 1 configuration is maintained through documented device configuration artifacts and engineering evidence.

However, the project does not implement a fully automated infrastructure configuration-management pipeline.

The following capabilities remain outside the current baseline:

- automated configuration deployment;
- automated configuration drift detection;
- automated compliance checking;
- centralized secrets management;
- automated backup and restore workflows.

These capabilities would become increasingly important in a larger production environment.

11.8 Security Validation Boundaries

The implemented security controls provide segmentation and access-control mechanisms appropriate to the Version 1 scope.

However, the validation does not constitute a complete penetration test or formal security assessment.

The project does not claim validation against:

- sophisticated Layer 2 attack tooling;
- advanced packet manipulation;
- real-world DHCP starvation tooling;
- MAC flooding at production scale;
- advanced firewall evasion;
- endpoint compromise;
- application-layer attacks.

The security validation therefore demonstrates the intended behavior of the implemented controls under the defined acceptance scenarios.

It does not establish that the network is resistant to every possible attack technique.

11.9 Failure Testing Boundaries

The failure and resilience tests in Chapter 9 intentionally cover selected failure domains.

The validated scenarios include:

- Core trunk failure;
- EtherChannel member failure;
- active Layer 3 Core failure.

These scenarios do not represent an exhaustive failure-injection program.

For example, Version 1 does not formally validate:

- simultaneous multi-device failures;
- complete access-switch failure;
- firewall hardware failure;
- complete power-domain failure;
- physical cabling failure across multiple paths;
- upstream ISP failure;
- extended-duration degraded operation.

The resulting resilience conclusions therefore apply only to the tested failure domains.

11.10 Documentation and Evidence Boundaries

The final engineering report intentionally summarizes the most important evidence rather than reproducing every configuration command and simulator output.

Detailed evidence remains associated with the project repository.

This separation provides two levels of documentation:

Final Engineering Report

```
|
+-- Design decisions
+-- Validation results
+-- Failure analysis
+-- Accepted limitations
|
v
```

Engineering Evidence

```

|
+-- Configuration artifacts
+-- Validation artifacts
+-- Device data
+-- Supporting diagrams

```

The report should therefore be interpreted together with the underlying project artifacts when detailed configuration-level verification is required.

11.11 Limitation Acceptance Matrix

The principal Version 1 limitations are summarized below.

Table 11.1: Version 1 Limitation Acceptance Matrix

Limitation	Classification	Disposition
No first-hop Layer 3 redundancy	Architectural	Accepted in V1; HSRP/VRRP deferred to V2
Packet Tracer ACL behavior / parser limitations	Simulation	Documented and accepted within validation scope
No physical hardware validation	Scope / Environment	Deferred to physical implementation phase
No production performance benchmark	Scope	Deferred until representative infrastructure is available
Limited centralized monitoring	Scope	Future operational enhancement
No automated configuration management	Scope	Future infrastructure automation enhancement
Limited failure-domain coverage	Scope	Expanded chaos testing deferred to future revision

11.12 Engineering Interpretation

The limitations documented in this chapter do not invalidate the Version 1 engineering baseline. Instead, they define the conditions under which the baseline should be interpreted.

The correct engineering statement is therefore:

Version 1 is validated within its defined architecture, scope, and simulation environment.

It is not equivalent to:

Version 1 is production-certified without further validation.

This distinction preserves technical accuracy while still allowing the project to function as a credible engineering portfolio baseline.

11.13 Future Resolution Path

The limitations identified in this chapter provide direct inputs to future engineering evolution.

The principal future improvements are:

1. Implement HSRP or VRRP for first-hop gateway redundancy.
2. Validate the architecture on physical or equivalent virtual Cisco infrastructure.
3. Introduce centralized logging and monitoring.
4. Add automated configuration backup and compliance validation.
5. Expand controlled failure testing.
6. Establish representative performance and capacity testing.

These improvements should be treated as controlled future changes rather than retroactively modifying the accepted Version 1 baseline.

11.14 Limitations Conclusion

The Version 1 Enterprise Network Infrastructure Blueprint has several documented boundaries, but none of these boundaries are hidden from the engineering baseline.

The most significant limitation is the absence of first-hop Layer 3 gateway redundancy. This limitation is explicitly tested, documented, and accepted.

Additional limitations arise from the Cisco Packet Tracer environment and from the intentionally bounded scope of the portfolio implementation.

By documenting these constraints explicitly, the project distinguishes between:

- what has been implemented;
- what has been validated;
- what has been intentionally deferred; and
- what requires a future production validation phase.

This establishes a transparent engineering boundary for Version 1 and provides the foundation for the as-built and acceptance documentation that follows.

Chapter 12 As-Built Configuration

This chapter documents the Version 1 as-built state of the Enterprise Network Infrastructure Blueprint.

Unlike the preceding architecture and design chapters, this chapter does not describe the intended future state of the network. It records the implemented baseline as validated within the Cisco Packet Tracer environment.

The as-built state therefore represents the configuration that was actually implemented, including approved design constraints and documented Version 1 limitations.

12.1 As-Built Baseline

The Version 1 implementation consists of the following principal infrastructure components:

- one active Layer 3 Core switch;
- one secondary Core switch;
- three access switches;
- one edge firewall;
- enterprise VLAN segmentation;
- Layer 3 SVI gateways;
- Layer 2 trunking;
- LACP EtherChannel;
- Rapid-PVST+ spanning-tree;
- access-layer security controls; and
- extended ACL policy enforcement.

The implementation was validated in Cisco Packet Tracer and represents the frozen Version 1 engineering baseline.

12.2 Device Inventory

The principal devices in the as-built topology are:

The inventory represents the Version 1 implementation rather than a hypothetical production expansion.

Table 12.1: Version 1 As-Built Device Inventory

Device ID	Role	Function
CORE-L3-01	Primary Core	L3 routing and SVI gateway
CORE-L3-02	Secondary Core	L2 redundancy / secondary path
ACC-F1-01	Access Switch	Access-layer connectivity
ACC-F2-01	Access Switch	Access-layer connectivity
ACC-F3-01	Access Switch	Access-layer connectivity
FW-EDGE-01	Edge Firewall	Perimeter / NAT function

12.3 VLAN and Subnet State

The as-built network uses the defined VLAN and subnet allocation.

The principal user and infrastructure VLANs are:

Table 12.2: Version 1 As-Built VLAN and Subnet Allocation

VLAN	Network	Primary Purpose
10	10.10.10.0/24	Development
20	10.10.20.0/24	Finance
30	10.10.30.0/24	HR
40	10.10.40.0/24	Management
50	10.10.50.0/24	Operations
60	10.10.60.0/24	Server Zone
70	10.10.70.0/24	Management / Transit
80	10.10.80.0/24	Infrastructure Services
90	10.10.90.0/24	Guest
999	–	Native / Black-hole VLAN

All principal user and infrastructure VLANs use the standardized /24 subnet allocation defined by the Version 1 network design.

12.4 Layer 3 Gateway State

The active Layer 3 gateway function is implemented on **CORE-L3-01**.

Representative SVI gateway addressing follows the VLAN allocation:

```

VLAN 10  -> 10.10.10.1/24
VLAN 20  -> 10.10.20.1/24
VLAN 30  -> 10.10.30.1/24
VLAN 40  -> 10.10.40.1/24
VLAN 50  -> 10.10.50.1/24
VLAN 60  -> 10.10.60.1/24
VLAN 70  -> 10.10.70.1/24
VLAN 80  -> 10.10.80.1/24
VLAN 90  -> 10.10.90.1/24

```

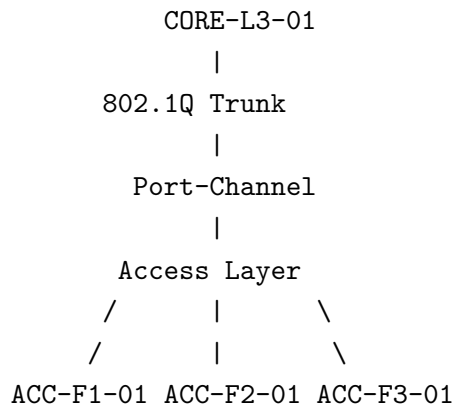
The Layer 3 Core therefore provides centralized inter-VLAN routing for the Version 1 topology.

12.5 Layer 2 State

The as-built Layer 2 architecture uses 802.1Q trunking between the Core and access-layer infrastructure.

The implementation also uses LACP EtherChannel where multiple physical links are aggregated into a logical Port-Channel.

The resulting logical model is:



Rapid-PVST+ provides loop prevention and maintains alternate Layer 2 paths where the topology contains redundant links.

12.6 Access-Layer State

The access switches provide endpoint connectivity and enforce the corresponding edge security controls.

The Version 1 access-layer baseline includes:

- access VLAN assignment;
- PortFast on endpoint-facing ports;
- BPDU Guard;
- DHCP Snooping;
- port-security controls;
- sticky MAC behavior where configured;
- approved trunk configuration on uplinks.

These controls collectively establish the intended Layer 2 security boundary between endpoint devices and the enterprise switching infrastructure.

12.7 Security Policy State

The as-built security state includes segmentation and access-control mechanisms implemented at the appropriate policy enforcement points.

The principal enforcement locations are:

Table 12.3: As-Built Security Enforcement Points

Component	Primary Security Function
FW-EDGE-01	Perimeter / NAT boundary
CORE-L3-01	L3 SVI routing and Extended ACL enforcement
Access Switches	PortFast, BPDU Guard, DHCP Snooping, Port Security

The security state described here represents the implemented Version 1 controls and does not imply the presence of security mechanisms that were deferred or excluded from the project scope.

12.8 As-Built Change Record

The documented endpoint change scenario **CHANGE-001** is represented in the as-built state.

The affected endpoint is:

DEV-PC-01

|

v

ACC-F2-01

Fa0/1

|

v

VLAN 10

|

v

10.10.10.0/24

This establishes the relationship between the change record and the final implemented network state.

The change is therefore not treated as an isolated historical event; its result is incorporated into the Version 1 as-built baseline.

12.9 Routing and Traffic State

The Version 1 routing model uses centralized Layer 3 forwarding through the active Core.

A representative internal traffic path is:

Endpoint

```

    |
    v
Access Switch
    |
    v
CORE-L3-01
    |
    +-----> Internal VLAN
    |
    +-----> Server VLAN
    |
    +-----> Management / Transit
    |
    v
FW-EDGE-01
    |
    v
External / Perimeter Network

```

The exact forwarding behavior is governed by the implemented routing table and security policy.

The as-built description therefore records the implemented forwarding architecture without introducing additional routing protocols or gateway redundancy not present in Version 1.

12.10 Resilience State

The as-built resilience model consists of:

- redundant Layer 2 paths;
- Rapid-PVST+ alternate-path behavior;
- LACP link aggregation;
- secondary Core participation at Layer 2.

The as-built state does **not** include first-hop Layer 3 gateway redundancy.

The active gateway remains:

CORE-L3-01

This distinction is intentionally preserved in the as-built documentation because it is one of the principal accepted Version 1 limitations.

12.11 Management and Infrastructure Addressing

The Version 1 implementation includes a dedicated management/transit segment represented by VLAN 70.

The principal Core management/transit addressing includes:

```
CORE-L3-01  
10.10.70.10/24
```

The management/transit segment provides the addressing required for the implemented Core and perimeter connectivity.

The as-built record does not infer additional management infrastructure that was not implemented in the Version 1 environment.

12.12 Configuration Evidence

The as-built state is supported by the project's configuration artifacts.

Representative verification commands include:

```
show ip interface brief  
show vlan brief  
show interfaces trunk  
show etherchannel summary  
show spanning-tree  
show ip route  
show access-lists  
show port-security  
show ip dhcp snooping
```

These commands provide configuration and operational evidence for the corresponding device state.

The complete command output is maintained in the project evidence rather than duplicated throughout the final report.

12.13 As-Built Versus Intended State

The Version 1 as-built state is intentionally compared with the approved design to identify deviations.

The comparison confirms that the principal difference between the intended Version 1 architecture and the final as-built state is the explicitly accepted absence of first-hop Layer 3 redundancy.

Table 12.4: As-Built State Assessment

Area	Approved V1 Design	As-Built State
VLAN segmentation	Defined VLAN architecture	Implemented according to VLAN plan
Layer 3 routing	Centralized Core routing	Implemented on CORE-L3-01
Layer 2 resilience	Rapid-PVST+ / alternate path	Implemented and validated
Link resilience	LACP EtherChannel	Implemented and validated
Security controls	Defined access-layer and L3 controls	Implemented within V1 scope
First-hop redundancy	Deferred to V2	Not implemented in V1

12.14 Baseline Freeze

Following implementation and validation, the Version 1 as-built state is treated as the frozen engineering baseline.

The freeze means that future modifications must be treated as controlled changes rather than silently incorporated into Version 1.

The baseline state can therefore be represented as:

Requirements

|

v

Approved Design

|

v

Implementation

|

v

Validation

|

v

As-Built State

|

v

V1 BASELINE

|

v

Controlled Future Change

This provides a clear separation between the accepted Version 1 state and future engineering evolution.

12.15 As-Built Conclusion

The Version 1 as-built record represents the network infrastructure actually implemented and validated in the Cisco Packet Tracer environment.

The baseline includes the approved VLAN architecture, centralized Layer 3 routing, Layer 2 resilience, LACP aggregation, access-layer security controls, perimeter enforcement, and the documented **CHANGE-001** endpoint placement.

The as-built state also explicitly preserves the accepted Version 1 first-hop gateway limitation.

Accordingly, this chapter provides the configuration-state reference against which the final acceptance decision can be evaluated.

Chapter 13 Acceptance

This chapter defines the formal acceptance status of the Version 1 Enterprise Network Infrastructure Blueprint.

Acceptance is based on the relationship between the approved requirements, implemented engineering design, validation results, documented limitations, and final as-built state.

The purpose of this chapter is to determine whether the Version 1 baseline can be accepted as a complete portfolio engineering baseline within its defined scope.

13.1 Acceptance Basis

The Version 1 acceptance decision is based on five evidence domains:

1. requirements traceability;
2. architecture and implementation consistency;
3. functional and security validation;
4. resilience and failure testing;
5. documented limitations and as-built state.

The acceptance decision therefore does not depend on a single successful connectivity test.

The baseline is considered accepted only when the implemented state is consistent with the defined engineering scope and all known limitations have been explicitly documented.

13.2 Acceptance Criteria

The Version 1 baseline is evaluated against the following acceptance criteria.

13.3 Requirement Acceptance

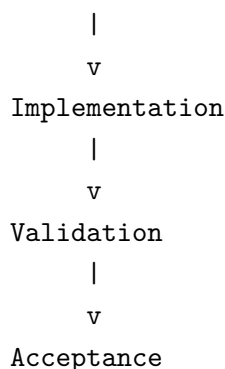
The requirements defined for Version 1 were traced through the engineering lifecycle.

The acceptance relationship can be represented as:

Requirement
|
v
Design Decision

Table 13.1: Version 1 Acceptance Criteria

Criterion	Status	Acceptance Basis
Requirements implemented	Accepted	Requirement traceability and implementation evidence
Network architecture implemented	Accepted	Architecture and as-built configuration
VLAN and Layer 3 segmentation	Accepted	Configuration and connectivity validation
Layer 2 resilience	Accepted	Spanning-tree and link-failure validation
LACP EtherChannel	Accepted	EtherChannel configuration and failure testing
Security controls	Accepted	Security validation scenarios
Failure scenarios	Accepted within scope	Defined resilience test scenarios
Known limitations	Accepted	Documented in Chapter 11
As-built state	Accepted	Version 1 implementation baseline



This traceability ensures that acceptance is based on implemented and validated engineering outcomes rather than design intent alone.

13.4 Functional Acceptance

The functional acceptance scope covers the principal network services and behaviors implemented in Version 1.

These include:

- VLAN segmentation;
- inter-VLAN Layer 3 routing;
- endpoint connectivity;
- trunk operation;

- EtherChannel operation;
- DHCP-related network behavior;
- perimeter connectivity; and
- defined access-control behavior.

The corresponding validation evidence is maintained in the project's validation artifacts and summarized in the Validation Matrix.

Within the defined test scope, the functional baseline is accepted.

13.5 Security Acceptance

Security acceptance is based on the implemented segmentation and access control mechanisms.

The Version 1 security baseline includes:

- VLAN-based segmentation;
- access-layer security controls;
- Layer 3 access-control policy;
- perimeter enforcement;
- restricted communication between defined security zones.

Acceptance of the security baseline means that the implemented controls performed according to the defined validation scenarios.

It does not constitute a claim of complete penetration-test certification or production security accreditation.

13.6 Resilience Acceptance

The resilience baseline is accepted within the failure domains explicitly tested during Version 1 validation.

The accepted resilience mechanisms include:

- redundant Layer 2 paths;
- Rapid-PVST+ convergence behavior;
- LACP EtherChannel resilience;
- secondary Core participation at Layer 2.

The active Layer 3 gateway remains:

CORE-L3-01

First-hop Layer 3 gateway redundancy is not part of the accepted Version 1 implementation.

This is consistent with the limitation documented in Chapter 11.

13.7 Limitation Acceptance

The Version 1 baseline contains known limitations that have been explicitly reviewed and accepted.

The principal limitation is the absence of first-hop Layer 3 gateway redundancy.

The acceptance decision does not require every potential future capability to be implemented.

Instead, the requirement is that excluded or deferred capabilities are clearly identified and do not contradict the approved Version 1 scope.

Therefore:

Known limitation $\neq$ unaccepted baseline

when the limitation is explicitly documented, assessed, and accepted within the project scope.

13.8 As-Built Acceptance

The final as-built state documented in Chapter 12 is used as the physical reference for the Version 1 acceptance decision.

The acceptance chain is:

Approved Design

|

v

Implemented Configuration

|

v

Validation Evidence

|

v

As-Built Baseline

|

v

ACCEPTANCE

The as-built state is considered consistent with the accepted Version 1 design, subject to the documented limitations.

13.9 Evidence Closure

The acceptance decision is supported by the project's engineering evidence.

The principal evidence categories are:

Table 13.2: Version 1 Acceptance Evidence

Evidence Category	Purpose
Requirement Traceability	Confirms requirement coverage
Architecture Documentation	Confirms intended engineering structure
Configuration Artifacts	Confirms implemented device state
Validation Report	Confirms functional test outcomes
Failure Test Evidence	Confirms tested resilience behavior
As-Built Record	Confirms final implemented baseline
Limitation Register	Confirms known constraints are documented

Together, these evidence categories provide the basis for closing the Version 1 engineering life-cycle.

13.10 Acceptance Decision

Based on the documented requirements, implementation evidence, validation results, failure testing, limitations, and as-built state, the Version 1 Enterprise Network Infrastructure Blueprint is:

ACCEPTED & FROZEN

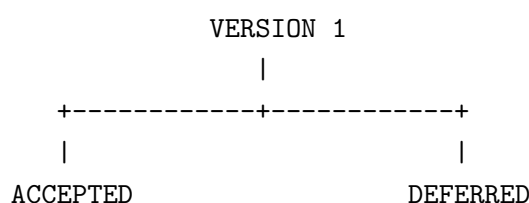
The acceptance status means that the Version 1 baseline satisfies the defined portfolio engineering scope and may be used as the reference state for subsequent controlled revisions.

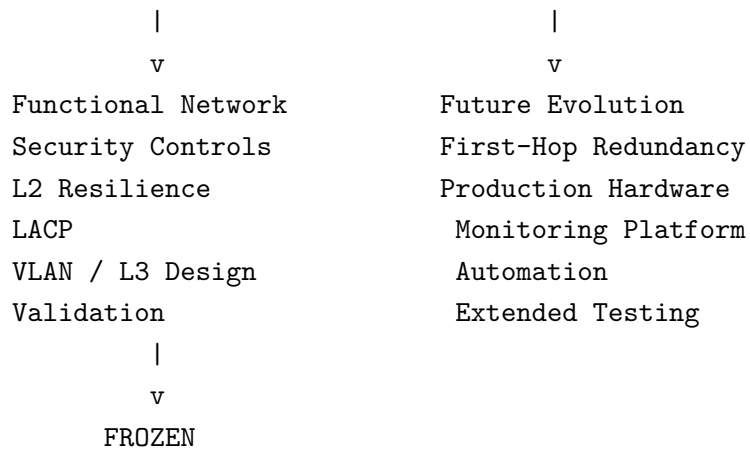
Acceptance does not imply that the architecture represents the final possible evolution of the network.

Future improvements must be introduced through controlled change and should receive their own implementation and validation evidence.

13.11 Acceptance Boundary

The acceptance boundary is defined as follows:





This boundary prevents future requirements from being retroactively treated as Version 1 acceptance failures.

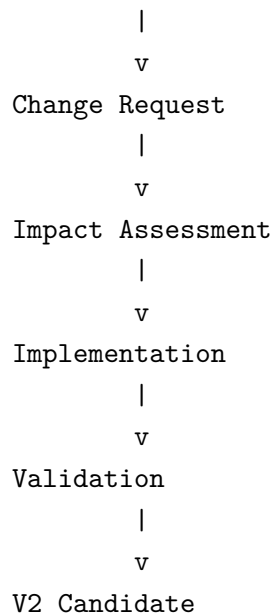
13.12 Baseline Freeze

Following acceptance, the Version 1 baseline is frozen.

Any future modification to the accepted state should be treated as a new controlled change.

The lifecycle therefore continues as:

V1 Accepted & Frozen



This maintains the relationship between the accepted baseline and future engineering evolution.

13.13 Final Acceptance Statement

The Version 1 Enterprise Network Infrastructure Blueprint has completed its defined engineering lifecycle.

Requirements were translated into an implemented network architecture, validated through defined functional, security, and resilience scenarios, and reconciled against the final as-built state.

Known limitations have been documented rather than concealed, with first-hop Layer 3 redundancy explicitly deferred to future evolution.

Accordingly, the Version 1 baseline is formally designated:

VERSION 1.0

ACCEPTED & FROZEN

The accepted baseline serves as the reference point for future engineering changes and subsequent versions of the Enterprise Network Infrastructure Blueprint.

Chapter 14 Future Evolution

This chapter defines the potential engineering evolution of the Enterprise Network Infrastructure Blueprint beyond the accepted Version 1 baseline.

The purpose of this chapter is to establish a controlled roadmap for future revisions rather than to modify the frozen Version 1 implementation.

Future capabilities are therefore derived from the limitations, operational boundaries, and engineering opportunities identified throughout the report.

14.1 Evolution Principles

Future revisions should preserve the engineering principles established by Version 1:

- changes must remain traceable to an engineering requirement;
- architectural changes must be evaluated before implementation;
- new capabilities must be validated before acceptance;
- security controls must remain consistent with the segmentation model;
- resilience improvements must address identified failure domains; and
- every accepted revision must produce an updated as-built baseline.

The objective is therefore evolutionary improvement rather than uncontrolled expansion of the network.

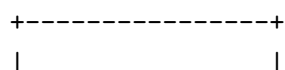
14.2 Priority 1 – First-Hop Layer 3 Redundancy

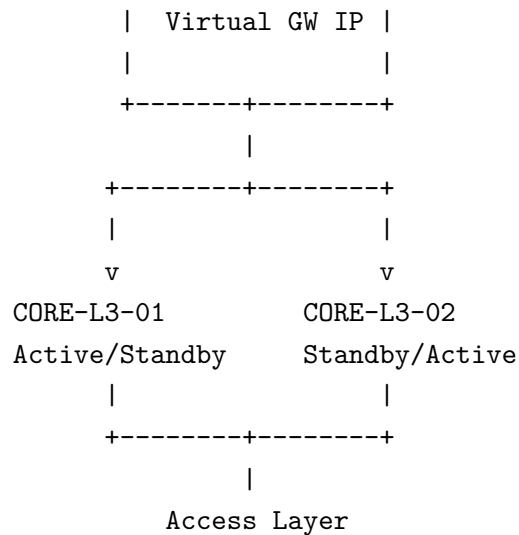
The highest-priority architectural enhancement is the introduction of first-hop Layer 3 gateway redundancy.

Version 1 currently uses **CORE-L3-01** as the active Layer 3 gateway. The secondary Core does not automatically assume the gateway role when the primary Core fails.

A future revision should evaluate an appropriate first-hop redundancy mechanism such as HSRP or VRRP.

The target architecture would be:





The implementation should include dedicated failure testing to demonstrate gateway continuity rather than simply confirming that the protocol is configured.

14.3 Priority 2 – Production Infrastructure Validation

The second major evolution is migration from simulation-based validation to validation on physical or production-equivalent infrastructure.

Future validation should evaluate:

- actual hardware forwarding behavior;
- interface and transceiver behavior;
- physical link failure;
- device reboot and recovery;
- routing and Layer 2 convergence;
- hardware-specific performance characteristics.

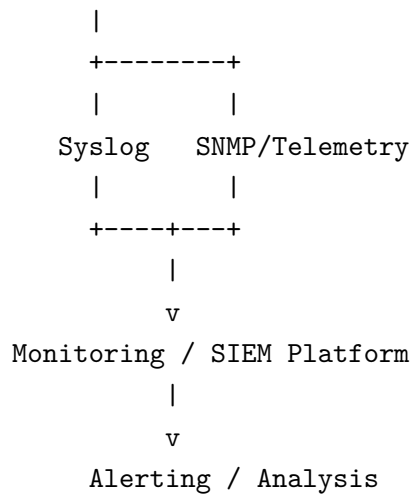
The existing Packet Tracer implementation should remain useful as a design and pre-deployment validation environment.

The physical validation phase would therefore complement rather than replace the existing engineering workflow.

14.4 Priority 3 – Centralized Monitoring

A future operational revision should introduce centralized network monitoring and logging.

The target operational model is:

Network Devices

Potential capabilities include:

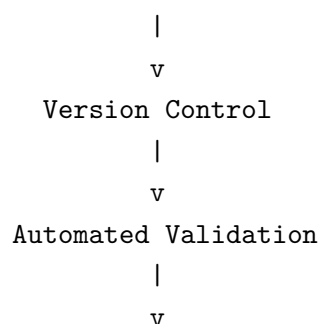
- centralized syslog collection;
- device health monitoring;
- interface utilization monitoring;
- link-state alerting;
- security-event correlation;
- configuration-change monitoring.

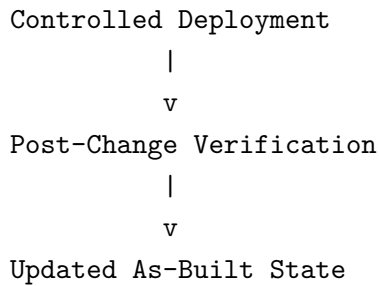
This would extend the current project from primarily infrastructure validation toward operational observability.

14.5 Priority 4 – Configuration Automation

Future versions may introduce infrastructure automation to reduce manual configuration and configuration drift.

A potential automation workflow is:

Approved Configuration



Potential capabilities include:

- automated configuration generation;
- configuration backup;
- configuration comparison;
- compliance checking;
- automated deployment;
- configuration drift detection.

Automation should only be introduced after the underlying configuration standards have been stabilized.

14.6 Priority 5 – Expanded Resilience Testing

The current failure-validation scope covers selected Layer 2, EtherChannel, and Core failure scenarios.

Future revisions should expand the failure domain coverage.

Potential scenarios include:

- complete access-switch failure;
- simultaneous link failures;
- perimeter-device failure;
- upstream connectivity failure;
- device reboot during active traffic;
- multiple simultaneous infrastructure failures.

The purpose is not to create failures arbitrarily, but to identify whether the architecture maintains an acceptable service state under defined failure conditions.

A future resilience model can therefore be represented as:

Single Failure

|

v

Component Failure

|

v

Multi-Link Failure

|

v

Multi-Device Failure

|

v

Service-Level Resilience

14.7 Priority 6 – Performance and Capacity Validation

Version 1 does not establish production performance benchmarks.

A future implementation should introduce representative traffic-generation and capacity testing.

Potential measurements include:

- throughput;
- packet loss;
- latency;
- jitter;
- interface utilization;
- convergence time;
- device resource utilization.

The results could then be compared against defined engineering capacity requirements.

This would extend the project from functional validation toward measurable performance engineering.

14.8 Priority 7 – Security Maturity

The current security architecture establishes segmentation and access control.

Future revisions could expand the security model toward operational detection and defense.

Potential improvements include:

- centralized security logging;

- IDS/IPS integration;
- enhanced network access control;
- security-event correlation;
- configuration compliance monitoring;
- periodic security validation.

These capabilities should be introduced according to identified risk rather than added solely for technological complexity.

14.9 Evolution Roadmap

The proposed evolution sequence is summarized below.

Table 14.1: Enterprise Network Evolution Roadmap

Priority	Evolution	Primary Objective
1	First-hop Layer 3 redundancy	Remove the current single active gateway dependency
2	Physical / production-equivalent validation	Verify behavior beyond simulation
3	Centralized monitoring and logging	Improve operational visibility
4	Configuration automation	Reduce manual changes and configuration drift
5	Expanded failure testing	Increase resilience assurance
6	Performance and capacity testing	Establish measurable operational limits
7	Security maturity enhancements	Extend prevention, visibility, and detection capabilities

The order is intentionally prioritized. First-hop redundancy has the highest architectural value because it directly addresses the principal availability limitation identified in Version 1. Monitoring, automation, performance engineering, and expanded security capabilities can then build on the stabilized architecture.

14.10 Version Evolution Model

Future revisions should follow the same engineering lifecycle used by Version 1.

The proposed model is:

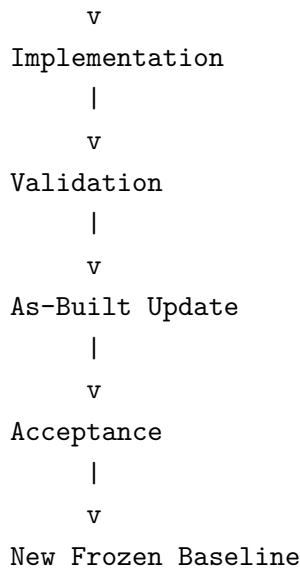
Requirement

|

v

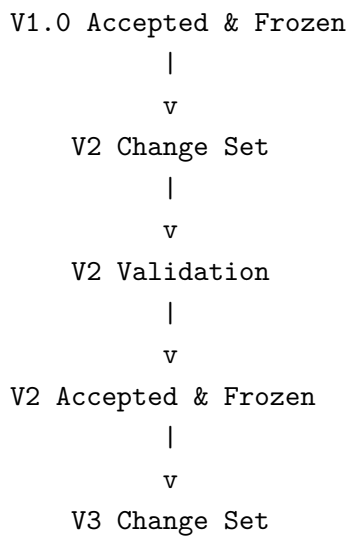
Architecture Change

|



Each version should therefore remain independently traceable.

For example:

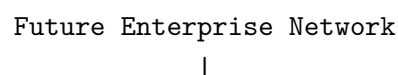


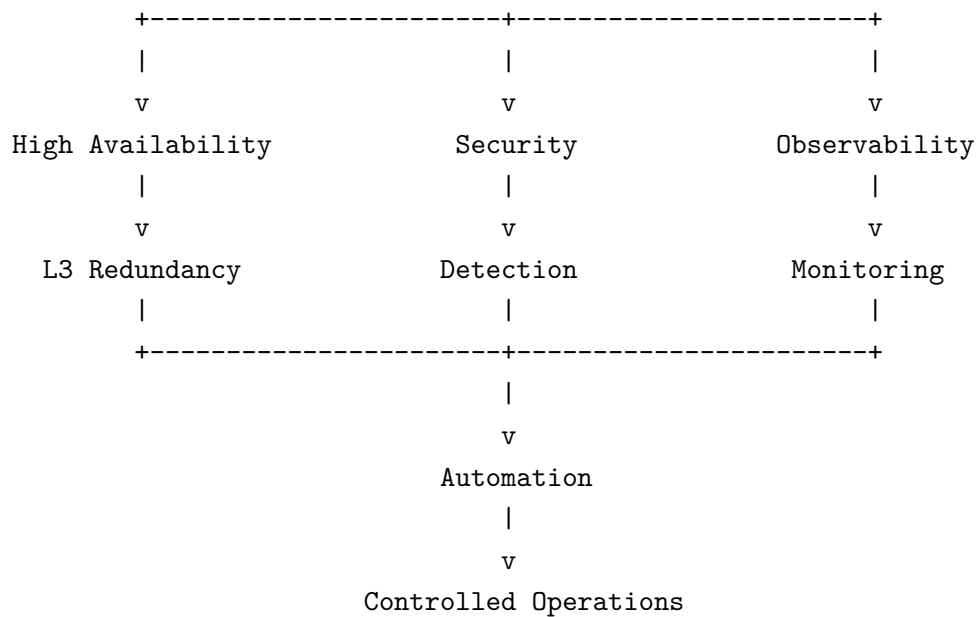
This prevents future modifications from silently changing the historical Version 1 engineering record.

14.11 Future Architecture Direction

The long-term direction of the network is toward increased availability, observability, automation, and operational maturity.

Conceptually:





This evolution does not require replacing the Version 1 architecture. Instead, the existing baseline provides the foundation upon which these capabilities can be introduced incrementally.

14.12 Future Change Governance

Every future enhancement should follow the change-management principles defined in Chapter 10.

A proposed future feature should therefore answer:

1. What requirement does the change address?
2. What existing components will be affected?
3. What security or availability impact is expected?
4. How will the change be validated?
5. What rollback mechanism is available?
6. What evidence will be captured?
7. How will the as-built baseline be updated?

This ensures that future evolution remains controlled and traceable.

14.13 Version 2 Candidate Scope

Based on the limitations identified in Version 1, the following capabilities are suitable candidates for a Version 2 engineering cycle:

- first-hop Layer 3 gateway redundancy;
- expanded resilience validation;
- physical or production-equivalent validation;
- centralized monitoring and logging;
- improved configuration-management practices.

These items are candidates rather than commitments. Their final inclusion should be determined by the requirements and acceptance criteria established for Version 2.

14.14 Final Engineering Outlook

Version 1 establishes a validated and documented enterprise network baseline.

Its primary value is not that every possible enterprise capability has already been implemented, but that the architecture, implementation, validation, limitations, and acceptance state are explicitly defined.

Future versions can therefore build on a known engineering foundation rather than restarting the design process.

The intended evolution is:

Design → Implement → Validate → Accept → Evolve

The Version 1 baseline is therefore considered complete within its defined scope, while providing a controlled foundation for future enterprise network maturity.

““latex

Appendices

Supporting Engineering Records

The following appendices provide supporting engineering records for the Version 1 baseline, including requirement traceability, validation evidence, and the as-built device inventory.

Chapter A Requirement Traceability

This appendix provides the consolidated traceability baseline for the Version 1 Enterprise Network Infrastructure Blueprint.

The purpose of this matrix is to maintain the golden thread from business and technical requirements through architecture, network design, implementation, and validation.

The canonical requirement identifiers are grouped into five requirement classes:

- FR — Functional Requirements;
- SR — Security Requirements;
- AR — Availability Requirements;
- SC — Scalability Requirements; and
- MG — Manageability Requirements.

These identifiers are used consistently throughout the engineering baseline.

A.1 Functional Requirements

Table A.1: Functional requirement traceability

ID	Requirement	Design Reference	Report Evidence
FR-01	Inter-Departmental Communication	Chapter 5 — Network Design	Chapter 8 — Validation
FR-02	Guest Internet Egress	Chapter 5 — Network Design	Chapter 8 — Validation
FR-03	Automated IP Allocation	Chapter 5 — Network Design	Chapter 8 — Validation
FR-04	Wireless Mobility Support	Chapter 5 — Network Design	Chapter 8 — Validation
FR-05	Secure Remote Administration	Chapter 5 — Network Design	Chapter 6 — Security

A.2 Security Requirements

Table A.2: Security requirement traceability

ID	Requirement	Design Reference	Report Evidence
SR-01	Subnet Logical Isolation	Chapter 6 — Security	Chapter 8 — Validation
SR-02	Confidential Asset Protection	Chapter 6 — Security	Chapter 8 — Validation
SR-03	Guest Untrusted Isolation	Chapter 6 — Security	Chapter 8 — Validation
SR-04	Rogue DHCP Mitigation	Chapter 6 — Security	Chapter 8 — Validation
SR-05	Layer 2 Loop Suppression	Chapter 6 — Security	Chapter 8 — Validation
SR-06	Control Plane Hardening	Chapter 6 — Security	Chapter 8 — Validation

A.3 Availability Requirements

Table A.3: Availability requirement traceability

ID	Requirement	Design Reference	Report Evidence
AR-01	Uplink Redundancy & Load Sharing	Chapter 5 — Network Design	Chapter 9 — Failure & Resilience
AR-02	Spanning-Tree & Path Redundancy	Chapter 5 — Network Design	Chapter 9 — Failure & Resilience
AR-03	Inline Power Delivery	Chapter 5 — Network Design	Chapter 8 — Validation

A.4 Scalability Requirements

Table A.4: Scalability requirement traceability

ID	Requirement	Design Reference	Report Evidence
SC-01	Modular Subnet Expansion	Chapter 5 — Network Design	Chapter 14 — Future Evolution
SC-02	Floor Density Replication	Chapter 5 — Network Design	Chapter 12 — As-Built
SC-03	IP Capacity Reservation	Chapter 5 — Network Design	Chapter 14 — Future Evolution

A.5 Manageability Requirements

Table A.5: Manageability requirement traceability

ID	Requirement	Design Reference	Report Evidence
MG-01	Standardized Config Codebases	Chapter 7 — Implementation	Chapter 10 — Change Management
MG-02	Deterministic Addressing Scheme	Chapter 5 — Network Design	Chapter 12 — As-Built
MG-03	Logical-to-Physical Traceability	Chapter 4 — Architecture	Chapter 12 — As-Built
MG-04	Disaster Recovery & Config Backups	Chapter 7 — Implementation	Chapter 10 — Change Management

A.6 Traceability Summary

The Version 1 baseline maintains traceability across the complete engineering lifecycle:

Requirements → Architecture → Network Design → Implementation → Validation

The requirement classes are intentionally separated so that functional, security, availability, scalability, and manageability objectives can be evaluated independently.

The matrix also establishes the relationship between requirements and the corresponding report chapters.

A requirement is therefore not considered complete merely because it appears in the requirements definition. Its implementation and validation status must also be considered.

Where the simulation environment imposes a limitation, such as the documented Packet Tracer limitation affecting SVI ACL enforcement, the requirement remains traceable while the limitation is recorded in the corresponding validation and limitation chapters.

This appendix therefore serves as the consolidated requirement traceability baseline for Version 1.

Chapter B Validation Matrix

This appendix provides the detailed Version 1 validation matrix for the Enterprise Network Infrastructure Blueprint. The matrix consolidates the engineering verification activities performed during the project and provides a direct relationship between the validation objective, expected behavior, observed result, and final status.

The matrix is intended to complement Chapter 8 rather than duplicate its narrative discussion.

B.1 Validation Methodology

Validation was performed against the implemented Cisco Packet Tracer baseline using configuration inspection, connectivity testing, Layer 2 verification, Layer 3 verification, security-control inspection, and controlled failure scenarios.

The validation approach follows the following sequence:

Requirement → Expected Behavior → Test / Inspection → Observed Result → Status

A result is classified as **PASS** when the observed behavior matches the defined engineering expectation. A result may be classified as **PASS WITH LIMITATION** when the intended design logic is validated but the simulation environment prevents complete hardware-level verification.

B.2 Layer 2 Validation

Table B.1: Layer 2 validation matrix

Test ID	Validation Objective	Expected Result	Observed Result	Status
L2-01	VLAN existence and segmentation	Required VLANs are present and assigned consistently	Required VLAN structure is present across the implemented switching baseline	PASS
L2-02	802.1Q trunking	Inter-switch trunk links carry the required VLANs	Trunking and VLAN propagation operate as designed	PASS

Test ID	Validation Objective	Expected Result	Observed Result	Status
L2-03	Rapid-PVST+ operation	Spanning-tree prevents Layer 2 loops while maintaining the intended forwarding topology	Rapid-PVST+ operates across the implemented switching topology	PASS
L2-04	LACP EtherChannel operation	Aggregated links operate as a logical resilient path	Port-channel members form the intended aggregated links	PASS
L2-05	LACP member failure	Traffic remains available when an individual member link fails	Remaining EtherChannel capacity maintains connectivity after member failure	PASS

B.3 Layer 3 Validation

Table B.2: Layer 3 validation matrix

Test ID	Validation Objective	Expected Result	Observed Result	Status
L3-01	SVI gateway operation	Configured VLAN interfaces provide the intended default gateways	Active SVIs provide Layer 3 gateway services for the implemented VLANs	PASS
L3-02	Inter-VLAN routing	Authorized traffic can traverse between required VLANs	Inter-VLAN routing operates through the configured Layer 3 gateway	PASS
L3-03	Default gateway reachability	Hosts can reach their configured gateway	Gateway reachability is maintained for the validated VLANs	PASS
L3-04	Routing path verification	Traffic follows the intended Layer 3 path	The implemented routing topology provides the expected forwarding path	PASS

B.4 Security Validation

Table B.3: Security validation matrix

Test ID	Validation Objective	Expected Result	Observed Result	Status
SEC-01	DHCP Snooping	Unauthorized DHCP server behavior is controlled on protected access segments	DHCP Snooping configuration is present in the security baseline	PASS
SEC-02	Port Security	Unauthorized endpoint MAC behavior is restricted at access ports	Sticky MAC / Port Security controls are included in the implementation baseline	PASS
SEC-03	BPDU Guard / PortFast	Access-edge ports are protected from unintended spanning-tree participation	BPDU Guard and PortFast controls are applied to the intended edge configuration	PASS
SEC-04	Guest isolation	Guest traffic remains isolated from protected internal network segments	Guest VLAN isolation is implemented as part of the security architecture	PASS
SEC-05	Extended ACL policy	Defined security policy is represented by the intended ACL logic	ACL policy logic is present; complete SVI enforcement is constrained by the Packet Tracer platform limitation	PASS WITH LIMI- TA- TION

B.5 Failure and Resilience Validation

Table B.4: Failure and resilience validation matrix

Test ID	Failure Scenario	Expected Result	Observed Result	Status
FAIL-01	Single EtherChannel member failure	Connectivity remains available through the surviving member links	Traffic remains available through the remaining aggregated path	PASS
FAIL-02	Spanning-tree path change	Rapid-PVST+ reconverges without creating a persistent Layer 2 loop	The switching topology maintains the intended resilient forwarding behavior	PASS
FAIL-03	Core path resilience	Redundant Layer 2 paths remain available when a primary path is unavailable	Layer 2 redundancy is maintained through the implemented topology	PASS
FAIL-04	Layer 3 gateway redundancy boundary	Active Layer 3 gateway failover is expected only when gateway redundancy is implemented	Version 1 intentionally operates with CORE-L3-01 as the sole active Layer 3 gateway; CORE-L3-02 provides Layer 2 redundancy only	LIMITATION

B.6 Validation Summary

The Version 1 validation baseline achieved an overall validation pass rate of **94.4%**. The result reflects successful verification of the principal Layer 2, Layer 3, security, and resilience objectives while preserving explicit limitations where the simulation environment or Version 1 architecture prevents complete validation.

Validation Category	Result
Layer 2 Infrastructure	PASS
Layer 3 Infrastructure	PASS
Security Controls	PASS WITH LIMITATION
Failure / Resilience	PASS WITH LIMITATION
Overall V1 Validation Base-line	94.4%

B.7 Validation Limitations

Two limitations are explicitly retained in the Version 1 baseline.

First, Packet Tracer does not provide a fully representative hardware environment for validating all ACL enforcement behavior. The ACL policy and intended security logic are therefore retained as design evidence, while hardware-level enforcement requires validation on representative physical or higher-fidelity equipment.

Second, Version 1 does not implement active Layer 3 gateway redundancy. CORE-L3-01 remains the active Layer 3 gateway while CORE-L3-02 provides Layer 2 redundancy. HSRP or VRRP is therefore treated as a future evolution item rather than a failed Version 1 requirement.

B.8 Acceptance Interpretation

The validation matrix supports the Version 1 acceptance decision. A “PASS WITH LIMITATION” result does not represent an undisclosed failure; instead, it identifies a known boundary between what was validated within the Cisco Packet Tracer environment and what requires a higher-fidelity implementation environment.

The resulting matrix therefore provides the evidence baseline for the acceptance status recorded in the final engineering report.

Chapter C Device Inventory

This appendix provides the Version 1 as-built device inventory for the Enterprise Network Infrastructure Blueprint.

The inventory establishes the relationship between the logical device identifier, device role, platform, and architectural layer. Device identifiers in this appendix follow the naming convention used by the Version 1 engineering baseline.

The inventory is intended to provide a concise reference for the implemented topology and to support logical-to-physical traceability.

C.1 Edge Devices

Table C.1: Version 1 edge device inventory

Device ID	Device Role	Platform	Architectural Function
FW-EDGE-01	Perimeter Firewall	Cisco ASA 5506-X	Edge security boundary and external traffic control
RTR-EDGE-01	Edge Router	Cisco ISR 4331	WAN / upstream routing boundary

C.2 Core Devices

Table C.2: Version 1 core device inventory

Device ID	Device Role	Platform	Architectural Function
CORE-L3-01	Primary Core / Layer 3 Switch	Cisco Catalyst 3650	Active Layer 3 gateway, SVI routing, and core switching
CORE-L3-02	Secondary Core / Layer 2 Switch	Cisco Catalyst 3650	Layer 2 redundancy and switching support

C.2.1 Layer 3 Gateway Boundary

Version 1 uses **CORE-L3-01** as the sole active Layer 3 gateway.

CORE-L3-01 provides the implemented SVI gateway functions and inter-VLAN routing services.

CORE-L3-02 participates in the switching architecture and provides Layer 2 redundancy, but does not operate as an active SVI gateway in the Version 1 baseline.

Active first-hop Layer 3 redundancy using HSRP or VRRP is therefore outside the Version 1 implementation scope and is reserved for a future evolution of the architecture.

C.3 Access Devices

Table C.3: Version 1 access device inventory

Device ID	Device Role	Platform	Architectural Function
ACC-F1-01	Floor 1 Access Switch	Cisco access switching platform	Endpoint access, VLAN assignment, and Layer 2 edge security
ACC-F2-01	Floor 2 Access Switch	Cisco access switching platform	Endpoint access, VLAN assignment, and Layer 2 edge security
ACC-F3-01	Floor 3 Access Switch	Cisco access switching platform	Endpoint access, VLAN assignment, and Layer 2 edge security

C.4 Inventory Summary

The Version 1 device inventory consists of the following logical device groups:

Device Group	Quantity
Edge Devices	2
Core Devices	2
Access Devices	3
Total Network Devices	7

The inventory reflects the Version 1 logical and physical topology used throughout the engineering report.

C.5 Device Naming Convention

The Version 1 naming convention is structured around the functional role and location of each device.

- FW-EDGE-01 identifies the primary perimeter firewall.
- RTR-EDGE-01 identifies the edge routing device.
- CORE-L3-01 identifies the primary Layer 3 core switch.
- CORE-L3-02 identifies the secondary core switch providing Layer 2 redundancy.
- ACC-F1-01, ACC-F2-01, and ACC-F3-01 identify the access switches assigned to Floors 1, 2, and 3 respectively.

This naming structure provides deterministic identification across the architecture diagrams, configuration artifacts, validation records, and as-built documentation.

C.6 Logical-to-Physical Traceability

The device inventory establishes the physical implementation reference for the logical architecture documented in Chapter 4 and the network design documented in Chapter 5.

The relationship can be summarized as:

Edge → Core → Access → Endpoints

The edge layer establishes the external network boundary, the core layer provides centralized switching and Layer 3 gateway services, and the access layer provides endpoint connectivity and Layer 2 security controls.

This inventory therefore serves as the as-built device reference for Version 1.

““